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H II region filling factors in NGC 628: Luminosity–size relation and connection with polycyclic aromatic hydrocarbon emission

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ABSTRACT

Aims. Understanding the internal structure of H II regions is fundamental for constraining star formation processes in galaxies. We investigated how the filling factor (FF) relates to luminosity, size, electron density, and Hα equivalent width (EW(Hα)) in H II regions. We also explored the association between star-forming regions and the polycyclic aromatic hydrocarbon (PAH)-to-dust emission, to assess their possible connection to the evolutionary stages of H II regions.
Methods. We investigated the FF in 622 H II regions in NGC 628 by combining 475 regions from SIGNALS and 147 from PHANGS-MUSE using a dedicated analysis pipeline. We derived the luminosity, emission line fluxes, radii, electron densities, and FF. We used PHANGS-JWST survey images to study the spatial relation between the FF and PAH-to-dust emission, quantified by the PAH-to-dust ratio R_{PAH} using James Webb Space Telescope/Mid-Infrared Instrument (JWST/MIRI) emission bands at 7.7, 11.3, and 21 μm.
Results. Higher Log(FF) and high EW(Hα) values are found in luminous regions, whereas more extended regions with lower EW(Hα) display lower Log(FF) values. We show that the H II radius definition significantly affects the $L_{H\alpha}$–R relation. Low-luminosity compact H II regions appear to reflect a transition from cluster-powered regions to nebulae that are ionized by single massive stars, with the transition occurring around $\log(L_{H\alpha}) \sim 37$ erg s^{-1}. The PAH-to-dust ratio R_{PAH} correlates with the volumetric Hα luminosity density, $L_{H\alpha}/R^3$, with a transitional range centered around $\log(L_{H\alpha}/R^3) \sim 32$ erg s^{-1}, corresponding to $\log(FF) \approx -4.4$. Regions with lower Log(FF) values exhibit higher R_{PAH} values, suggesting less efficient PAH processing in more porous structures.
Conclusions. These results can be consistent with an evolutionary scenario in which the FF decreases with stellar cluster age in giant H II regions as they evolve toward fainter and more spatially extended states. The volumetric Hα luminosity density, $L_{H\alpha}/R^3$, provides a useful representation that reduces the covariance between $L_{H\alpha}$ and R induced by different region definition methods, enabling more consistent cross-catalog comparisons in segmentation approaches.

Key words. dust, extinction – HII regions – galaxies: star formation

1. Introduction

Star-forming galaxies actively form stars over long periods of time. Studies of galaxy star formation histories have shown that galaxies have been forming stars over billions of years (Madau & Dickinson 2014). Massive and dense molecular clouds are the birthplaces of regions hosting hot massive stars (O- and early B-type stars). Feedback from these young stars on the surrounding interstellar medium (ISM) occurs in the form of stellar winds, ionizing and non-ionizing radiation, and eventually, supernova explosions. These feedback mechanisms locally enrich the ISM and are thought to regulate and, in some cases, trigger subsequent star formation episodes (Fichtner et al. 2024). Photoionization plays a key role in this process. Ionizing photons interact with the surrounding gas, giving rise to H II regions that emit characteristic nebular emission line spectra. By

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analyzing these emission lines, we can infer the physical and chemical properties of the gas. Furthermore, examination of emission line profiles provides insight into the kinematics of the gas, including expansion and turbulence along the line of sight.

These H II regions produce characteristic optical emission, such as Hα, which is commonly used as a tracer of recent star formation on timescales shorter than 10 Myr (Kennicutt & Evans 2012). Observations reveal that H II regions exhibit complex internal structures, with dense zones embedded within lower-density gas and spanning a wide range of densities and temperatures (e.g., Castaneda et al. 1992). Several models have been proposed to describe these inhomogeneities. The first studies to characterize internal density fluctuations were conducted by Osterbrock & Flather (1959). They proposed that as a first approximation, H II regions can be modeled as spheres filled with low-density gas containing small optically thin high-density clumps or condensations. These clumps form parsec-scale substructures that are coherent in position-velocity space (Williams et al. 1999), and electron density variations within individual regions can be traced using density-sensitive emission-line ratios such as [S II].

In a spherical nebula with uniform physical conditions within the ionized region, the Hα luminosity is obtained by integrating the volume emissivity, $j_{H\alpha}(\mathbf{r})$, over the emitting volume,

$$L(H\alpha) = \int_V j_{H\alpha}(\mathbf{r})\, dV = \int_V h\nu_{H\alpha}\, \alpha^{\rm eff}_{H\alpha}(T)\, n_e(\mathbf{r})\, n_p(\mathbf{r})\, dV, \quad (1)$$

where n_e is the electron density, n_p is the proton density, $\alpha^{eff}_{H\alpha}(T_e)$ is the $H\alpha$ effective recombination coefficient, and $h\nu_{H\alpha}$ is the photon energy. In a fully ionized hydrogen nebula, $n_e \simeq n_p$ and when we consider a fully spherical volume uniformly filled with ionized gas of constant electron density, that is, with a filling factor, FF, as FF = 1, and ionization-bounded conditions, the Hα luminosity can be written as

$$L(H\alpha) = j_{H\alpha}(T_e)\, n^2_{e,\rm hom}\, V. \quad (2)$$

When we define $n^2_{e,\rm hom}$ as the root-mean-square ($\langle n^2_e\rangle$; rms) the electron density averaged over the entire geometrical volume, the electron density in the homogeneous case or rms can be derived as

$$\langle n^2_e\rangle = \frac{1}{V}\int_V n^2_{e,\rm hom}(\mathbf{r})\, dV \quad (3)$$

$$L(\mathrm{H}\alpha) = \langle n^2_e\rangle\, j_{H\alpha}(T_e)\, V \quad \Rightarrow \quad \langle n^2_e\rangle = n^2_{e,\rm hom} = \frac{L(\mathrm{H}\alpha)}{j_{H\alpha}(T_e)\, V}. \quad (4)$$

In real H II regions the ionized gas does not fill the entire geometrical volume. Instead, the emission arises from dense clumps embedded in a more diffuse medium. In this case, an $FF < 1$ must be introduced to account for the fraction of the total volume actually occupied by gas with the local electron density, $n^2_{e,\rm loc}$. The Hα luminosity then becomes

$$L(\mathrm{H}\alpha) = FF\, n^2_{e,\rm loc}\, j_{H\alpha}(T_e)\, V \quad \Rightarrow \quad n^2_{e,\rm loc} = \frac{L(\mathrm{H}\alpha)}{FF\, j_{H\alpha}(T_e)\, V}. \quad (5)$$

The quantity $n_{e,\rm loc}$ represents the local electron density within the gas clumps. It is typically estimated from optical spectroscopy by density-sensitive collisionally excited line ratios such as the doublet [S II]$\lambda\lambda$ 6716 /6731. This is the density of the gas that produces the emission lines. By contrast, the density derived under the homogeneous assumption, $n_{e,\rm hom}$, corresponds to the mean luminosity-weighted or root-mean-square (rms), that is the electron density averaged over the entire geometrical volume, derived from Hα, Hβ, or radio continuum emission. Then, from Eqs. (4) and (5) we obtain

$$n^2_{e,\rm hom} = FF\, n^2_{e,\rm loc} \quad \Rightarrow \quad \mathrm{FF} = \left[\frac{\langle n_e\rangle}{n_e}\right]^2. \quad (6)$$

Equation (6) is based on the assumption of a two-phase medium, as discussed above, in which the dense clumps are characterized by a uniform density, while the inter-clump medium contributes negligibly to the observed emission. Thus, the relation between both quantities is the FF, or the fraction of the total geometrical volume occupied by ionized gas with the local density (Osterbrock & Flather 1959).

In the following, we denote the local electron density $n_{e,\rm loc}$ simply as n_e, while the rms electron density, $\langle n_e\rangle$, denotes the $n_{e,\rm hom}$. By rearranging the variables in Eq. (5), we express the FF in terms of the other independent variables,

$$\log(\mathrm{FF}) = \log(\mathrm{L_{H\alpha}}) - \log(\mathrm{R^3}) - 2\log(\mathrm{n_e}) - 31.6. \quad (7)$$

where 31.6 arises from combining the Hα emissivity at $T_e = 10^4$ K, the geometrical factor $4\pi/3$ for a spherical volume, and the conversion from parsecs to centimeters.

The difference between these two measurements can reach nearly two orders of magnitude. It quantifies the degree of internal inhomogeneity, or porosity, of the ionized gas within each individual nebula (Kennicutt 1984; Pathak et al. 2025). Subsequent models incorporating more realistic physical conditions were developed to better reproduce observations of nearby galactic H II regions, such as the Orion nebula (e.g. Rubin et al. 1991; O'Dell et al. 2008; Simón-Díaz et al. 2011), and to address radiative transfer effects and the fraction of Lyman photons that escape from the nebula (e.g. Giammanco et al. 2004). More recently, computational approaches based on probabilistic density distributions have been developed to model internal density fluctuations within H II regions, and explore potential correlations between density, temperature, and the FF (Bergerud et al. 2019).

However, from the point of view of H II region models when applied to spatially resolved extragalactic H II regions, the FF should be interpreted as a global parameter that includes at least three physically distinct components. These are i) a clumping factor, representing small-scale density inhomogeneities within the ionized gas volume and analogous to the FF used in photoionization models, ff; ii) a covering factor, corresponding to the fraction of the ionizing photon field intercepted by the nebula ($\Omega/4\pi$) as seen from the ionizing source; and iii) a geometric or cavity factor, describing the morphology of the region through the ratio of the volume of the sphere defined by the outer radius and the volume of the shell actually occupied by the ionized gas. In the case of a filled sphere, this factor approaches unity, whereas it is much lower than unity for thin shell-like geometries,

$$FF = ff \times \frac{\Omega}{4\pi} \times \left(\frac{R_{out} - R_{cavity}}{R_{out}}\right)^3. \quad (8)$$

The FF is therefore not only linked to the relative distribution of dense clumps and the physical processes that enabled their survival, it is also tightly connected to the spatial scale over which electron density variations are measured, and thus,

to the ions used as tracers. Therefore, it is crucial to understand the scale of electron density variations for unraveling the mechanisms driving star formation. These variations are important for interpreting the gas structure and the physical processes regulating star formation because they directly affect the inferred FF (Mathis & Wood 2005).

A comprehensive understanding of the FF remains challenging, largely due to the historical lack of observations at sufficiently high spatial and spectral resolution. Instrumental limitations have long hindered us to resolve fine-scale nebular structures and accurately characterize density variations. However, the advent of new telescopes and integral-field spectrographs has enabled studies to routinely achieve sensitivities and angular resolutions well below ∼1″. In NGC 628, the Physics at High Angular Resolution in Nearby Galaxies Survey – Multi Unit Spectroscopic Explorer (PHANGS-MUSE; Leroy et al. (2021); Emsellem et al. (2022)) reaches a spatial resolution of ∼ 43.78 pc, while the Star formation, Ionized Gas, and Nebular Abundances Legacy Survey (SIGNALS; Rousseau-Nepton et al. (2019)) achieves ∼35 pc, enabling detailed characterization of H II region properties and robust estimates of the FF in large samples.

Only a limited number of studies have investigated the FF in extragalactic H II regions. Hunt & Hirashita (2009) linked the FF values in the range of $\sim 10^{-3}$–10^{-1} to dust optical depth by examining size-density relations and incorporating dust extinction effects in H II regions. They found that a decreasing FF corresponds to an increasing optical path length ($\tau \propto \delta^{-1/6}$; a change of 1 dex in the FF resulted in a tripling of the optical path). They concluded that as the FF increases, the volume occupied by clumps is diminished, while the space between them expands. Kennicutt (1984) reported a somewhat narrow FF range of (10^{-2}–10^{-1}) for Galactic and supergiant extragalactic H II regions. More recently, Cedrés et al. (2013) analyzed 58 H II regions in NGC 6946 and extended the FF range down to $\sim 10^{-3}$ for regions smaller than ∼50 pc, highlighting a dependence on galactocentric radius. A conservative lower limit of log(FF) ∼ −7 can be considered when extrapolations toward more diffuse ionized structures, consistent with earlier extragalactic studies (e.g., Cedrés et al. 2013; Hunt & Hirashita 2009). At larger scales, Reddy et al. (2023) showed that high FF values in star-forming galaxies correlate with elevated star formation rates and surface densities, and significantly affect the ionization parameter. More recently, Zurita et al. (2026) derived the rms electron density, local electron density, and FF for H II regions in NGC 2403 and NGC 628. These two galaxies follow a similar size–density relation, $\langle n_e \rangle_{\rm rms} \propto R^{-0.3}$ for $R \lesssim 50$ pc, with no clear dependence of n_e or FF on galactocentric radius. The authors also showed that rms densities and the slope of the $L_{H\alpha}$–R relation are sensitive to the cataloging method, highlighting the need for homogeneous approaches when comparing H II region properties in galaxies. Despite these efforts, the spatial distribution of FF within galaxies, its dependence on local environment, and its potential association with filamentary dust structures remain poorly constrained (Smith et al. 2015).

The star formation activity is distributed across the entire galaxy (e.g., Ragan et al. 2018; Sun et al. 2024; Querejeta et al. 2024). It is intertwined with an extended network of filamentary dust and gas structures that span scales from tens of parsecs to several kiloparsecs (Kumar et al. 2020; Pineda et al. 2023). Studies examining the independent evolution of star-forming regions and dust filaments have not found a one-to-one correspondence at all spatial scales (Louie et al. 2013). Using James Webb Space Telescope/Mid-InfraRed Instrument (JWST/MIRI) observations combined with PHANGS-Hα maps of NGC 628, Kim et al. (2023) showed spatial offsets between different star formation tracers and molecular gas on ∼100 pc scale. This is indicative of distinct evolutionary stages in the star formation cycle. Similarly, Thilker et al. (2023), used mid-infrared (MIR) flux measurements along with Hα luminosity values of H II regions from the catalog of Groves et al. (2023) to report that at least 75% of star formation is associated with the filament structures on ∼25 pc scales.

At sub-kiloparsec scales, corresponding to the typical scale of classic H II regions, dust and polycyclic aromatic hydrocarbons (PAHs) coexist with ionized gas and emit strongly in mid-infrared (e.g., Tielens 2013; Shipley et al. 2016; Oey et al. 2017; Herrero et al. 2022). PAH emission primarily arises from the photoevaporation or disintegration of larger carbonaceous grains by ultraviolet radiation from massive young stars, and it represents an important component of interstellar dust. PAHs undergo several processes, such as UV photodissociation, ion bombardment, and accretion reactions involving H, C+, and O. Of these, UV photodissociation exhibits the highest reaction rate (Omont & Bettinger 2021). De-excitation through vibrational modes gives rise to the Aromatic Infrared Bands (AIBs) at 3.3, 6.2, 7.7, 8.6, 11.3, and 12.7 μm, where the band at 7.7 μm traces predominantly ionized PAH, and the 11.3 μm band traces neutral PAHs (Rapacioli et al. 2004).

Early observations at 8 μm with *Spitzer* Space Telescope/IRAC showed that mid-infrared emission is spatially concentrated around H II regions (Phillips & Ramos-Larios 2008). The advent of JWST/MIRI now enables high-resolution infrared studies of PAH emission, opening new opportunities to investigate PAH evolution as a function of local H II region properties. Several studies indicate that PAHs become more ionized and smaller with increasing star formation surface density (e.g., Thilker et al. 2007; Egorov et al. 2023a; Rigopoulou et al. 2024; Ujjwal et al. 2024; Baron et al. 2025), likely due to enhanced destruction by hard radiation fields. The PAH abundance relative to the total dust population can be assessed by comparing PAH-dominated MIR bands (7.7 μm and 11.3 μm) with the mid-infrared emission from warm dust at 21 μm. This approach is related to the q_{PAH} parameter, defined as the fraction of total dust mass in PAHs containing fewer than 10^3 carbon atoms (Draine & Li 2007). The parameter q_{PAH} modulates the spectral shape of the infrared emission, preferentially affecting PAH-dominated bands (Draine et al. 2021). In this framework, the JWST F2100W band provides a reliable proxy for the total infrared emission (Chastenet et al. 2023), while the F770W and the F1130W bands are selectively sensitive to PAH emission. Accordingly, we used an empirical tracer of the q_{PAH} fraction defined as $R_{PAH} = \frac{F_{770W}+F_{1130W}}{F_{2100W}}$.

Recent observations show that the PAH fraction decreases significantly within H II regions, consistent with the efficient destruction of small grains in ionized environments (e.g., Egorov et al. 2023b; Chastenet et al. 2023; Sutter et al. 2024; Dale et al. 2025; Egorov et al. 2025). Moreover, the PAH fraction is negatively correlated with the Hα equivalent width of H II regions (Egorov et al. 2023a), suggesting a link with evolutionary stage.

We sought to obtain the most extensive and complete census of the FF to date within a single galaxy. We focused our analysis on the nearby spiral galaxy NGC 628 (M74), an almost face-on grand-design galaxy (class 9 in the classification of Elmegreen & Elmegreen (1987)). The global properties of NGC 628 are summarized in Table 1. We investigated its relation with the physical and chemical properties of H II regions and assessed its connection with PAH-to-dust emission.

Table 1. Global properties of NGC 628.

Property	Units	Value	References
RA (J2000)	hh:mm:ss	01:36:41.747	NED
Dec (J2000)	dd:mm:ss	+15:47:01.18	NED
Distance	Mpc	9.84	Anand et al. (2021)
Inclination[a]	°	8.9	Lang et al. (2020) Berg et al. (2013)
R_{25}	kpc	10.95	(Makarov et al. 2014)

Notes. *a* Inclination adopted in SIGNALS is 21° (Rousseau-Nepton et al. 2018).

This paper is structured as follows. Section 2 presents the data used. Section 3 describes the analysis procedures and the H II region selection procedures. Section 4 examines the dependence of the Hα luminosity-size relation on the adopted region definition. Section 5 introduces the derived physical parameters. Section 6 presents the main results on the FF and its relation with PAH emission. Section 7 compares our findings with previous studies. Finally, Section 8 summarizes our conclusions and outlines prospects for future work.

2. Data sample

We used public MUSE (Multi Unit Spectroscopic Explorer at the Very Large Telescope, VLT; ESO Paranal Observatory, Chile; (Bacon et al. 2010)) data from the PHANGS survey[1] (Emsellem et al. 2022), which observed spectroscopically 19 nearby galaxies to study the star formation history and the ISM with MUSE, complemented by observations from the SIGNALS survey[2] (Rousseau-Nepton et al. 2019), a program designed to study star formation and ionized regions in ∼35 local galaxies.

We focused on NGC 628 which has also been observed by the Spectro-Imageur à Transformée de Fourier pour l'Étude en Long et en Large des raies d'Émission (SITELLE) at the Canada-France-Hawaii telescope (CFHT) (Drissen et al. 2019). We selected our H II region candidates from the published SIGNALS survey and identified PHANGS-MUSE regions by using Hα images from the PHANGS-MUSE survey. Table 1 outlines the main properties of this galaxy.

We further combined these data with PHANGS-JWST observations (PHANGS-JWST; Lee et al. (2023)) to examine the relation between H II regions and PAH-to-dust emission.

2.1. SIGNALS survey

The first subset of our H II region sample was drawn from the catalog compiled by Rousseau-Nepton et al. (2018). NGC 628 was observed using the imaging Fourier transform spectrometer SITELLE. This instrument operates in the optical regime with a field-of-view (FoV) of 11 ′ × 11 ′, corresponding to 121 ′ (∼992 kpc^2 at the distance of NGC 628), providing full spatial coverage of the galaxy in a single pointing. The mean pixel scale is 0.321″× 0.321″, and the spatial resolution is ∼ 35 pc corresponding to a full width at half maximum (FWHM) of ∼0.8 ″ at the distance of 9.84 Mpc. The spectral coverage is provided by three filters: SN1 (364 nm to 385 nm), SN2 (484 nm to 512 nm), and SN3 (648 nm to 686 nm), with resolving powers R = 600, 600 and 1800, respectively. Further details on the observational setup can be found in Rousseau-Nepton et al. (2018).

The resulting data cube was processed following several steps using SITELLE's dedicated reduction pipeline ORBS (Martin et al. 2012), the spectral extraction engine ORCS (Martin et al. 2015) and an H II region detection algorithm described in Rousseau-Nepton et al. (2018). We summarize the main steps here (i) subtraction of the stellar continuum using a reference spectrum representative of the old underlying stellar population, extracted from the peak of the galaxy continuum and scaled to the local continuum level of each region; (ii) identification of emission peaks by comparing the intensity of adjacent pixels and applying a detection threshold; (iii) definition of the zone of influence (ZoI) around each emission peak, up to a maximum radius of 425 pc; (iv) determination of the H II region boundaries in $H\alpha$ by identifying the radius at which the slope of the total flux in the flux-distance diagram decreases by less than 2 %; and v) correction for the local diffuse ionized gas (DIG) background using the median intensity of an annulus surrounding the outer boundary of each region. Dust extinction was corrected using the Cardelli et al. (1989) extinction law and the theoretical $F(H\alpha)/F(H\beta) = 2.87$, assuming Case B recombination, an electron temperature of 10^4 K and low-density conditions. The final published SIGNALS catalog contains 4285 H II region candidates and is publicly available.

2.2. PHANGS-MUSE survey

The PHANGS-MUSE mosaic covers the central star-forming disk of NGC 628, spanning a total area of ∼89 kpc^2 assembled from multiple IFU pointings (Li et al. 2024). The observations were obtained using the MUSE wide-field mode and combined into a mosaicked data cube. To ensure uniform image quality, all cubes were convolved to match the largest point spread function (PSF) measured across the observation, resulting in a PSF-homogenized dataset for NGC 628 with a final FWHM of 0.92″. These products, referred to as the "copt" cubes, were reduced using the tailored pipeline described in Emsellem et al. (2022). The native MUSE pixel scale in wide-field mode is 0.2″. With a PSF FWHM of 0.92″, corresponding to a physical resolution of 43.8 pc, the MUSE resolution is slightly coarser than, but comparable to, the SIGNALS resolution (Santoro et al. 2022). The spectral range covered by MUSE spans from 4750 Å to 9300 Å.

2.3. Additional data

We compared our measurements with ancillary data from three published catalogs of H II regions in NGC 628 based on different segmentation methods: Groves et al. (2023); Congiu et al. (2023) using PHANGS-MUSE data, Barnes et al. (2026) with PHANGS-HST data, and Zurita et al. (2026) with ground-based narrow-band Hα imaging at a lower spatial resolution than the PHANGS-MUSE datasets.

The catalog of Zurita et al. (2026) for NGC 628 does not provide measurements of the [S II] $\lambda\lambda$6716, 6731 emission line fluxes, making it impossible to determine electron densities for individual regions. Because estimates of n_e are required to compute the FF, this catalog cannot be included in the FF scaling analysis. It is therefore considered only in the analysis of the $L_{H\alpha}$–R relation presented in Sect. 4, which requires only luminosity and radius measurements.

[1] https://www.phangs.org
[2] https://signal-survey.org

These catalogs rely on distinct segmentation and boundary definition strategies, including variations in the treatment of the DIG and spatial resolution. As a result, systematic differences in the derived sizes and luminosities of the identified regions may affect scaling relations such as the *L–R* relation we explored.

Finally, to extend our analysis into the infrared regime and trace PAH and thermal dust emissions, we used publicly available MIRI observations of NGC 628 from the PHANGS-JWST survey. Details on the observing strategy and imaging post-processing are provided in Lee et al. (2022) and Williams et al. (2024). These data allow us to probe different components of the star-forming environment and obtain a comprehensive view of the birthplaces of H II regions in this galaxy. We employed the MIRI F770W, F1130W, and F2100W filters. The angular resolution achieved is 0.238″ for F770W (corresponding to ∼11.3 pc), 0.36″ for F1130W (∼17 pc), and 0.651″ for F2100W (∼ 30 pc). The images correspond to the v1p0p1 data release and were processed with the standard JWST Science Calibration Pipeline (Bushouse et al. 2024; Williams et al. 2024), yielding maps in units of MJy/sr. We evaluated the background level in the full FoV of each image and identified a small residual baseline in the F2100W band. However, this offset was not subtracted, as the signal-to-noise ratio of the residual relative to the background dispersion is below 3, indicating that it is not statistically significant. All the infrared images were convolved to the SIGNALS PSF of 0.8″ or the PHANGS-MUSE PSF of 0.92″ using the convolution kernels and procedures described in Aniano et al. (2011), ensuring a consistent spatial resolution across all datasets. The F770W filter primarily traces ionized PAH emission at 7.7 µm, the F1130W filter is sensitive to neutral PAHs at 11.3 µm, and the F2100W filter traces warm dust emission at 21 µm (e.g., Galliano et al. 2018; Rigopoulou et al. 2021).

3. H II region identification and catalog construction

3.1. Definition of the PHANGS-MUSE nebular region

We developed a modular analysis pipeline to identify and characterize nebular regions in nearby galaxies, incorporating a novel treatment of local DIG and morphological complexity. Regions are initially segmented using HIIDENTIFY (Easeman et al. 2023) on the Hα emission maps. We applied a signal-to-noise (S/N) ratio threshold of S/N > 3 in order to suppress noise-dominated spaxels. The code applies an iterative growth around the brightest pixel flux and incorporates neighboring pixels whose flux exceeds a user-defined background threshold. In NGC 628, we disable this background lower limit so that the flux growth is driven solely by local emission. The growth is halted when the fraction of discarded pixels in the surrounding annulus reaches 80%. To refine the region selection, we imposed two additional constraints: (i) we required at least eight contiguous pixels, and (ii) the spatial resolution was set as the minimum separation defined by the FWHM. The algorithm does not impose constraints on region geometry, allowing for complex and irregular morphologies. Spaxels associated with DIG are not masked at this stage, as their treatment is deferred to subsequent analysis steps. A total of 4830 regions were initially identified. To consolidate all spaxels belonging to the same region and mitigate isolated pixel-level artifacts in the Hα map, we applied a topological cleaning procedure based on connected-component analysis.

Each region is subsequently analyzed in flux–distance space, where radial profiles are fit with a pseudo-Voigt function. The fitted parameters include the peak intensity, background offset, standard deviation (*sigma*), Gaussian-to-Lorentzian ratio (*alpha*), and the coefficient of determination, R^2_{fit}, adopted as a goodness-of-fit indicator. These profiles are used to define the cutoff radius and derive morphological properties.

The spatial extent of each region of interest (ROI) or potential H II region spectrum is defined using a hierarchical decision tree based on the statistical robustness of the radial profile fit, R^2_{fit}, and on the presence of extended or secondary emission components. Rather than adopting a fixed aperture, we combine analytical profile fitting, dispersion information, morphological constraints, and conservative flux thresholds to determine the final integration radius (typically in the order of $\sqrt{2}\sigma_{Pvoigt}$), for each region and avoid contamination from neighboring structures.

Additionally, the final radii are selected to exceed the instrumental resolution. This is achieved by adopting the FWHM of the PSF as the threshold. A full description of the method is presented in a forthcoming paper (Aragüete Riesco et al., in prep.). H II region geometrical parameters, including centroid, position, semi-major and semi-minor axis lengths, eccentricity, solidity, and orientation, are derived from minimum bounding ellipse fits.

The galactic DIG subtraction is performed by characterizing the galactic diffuse emission within adaptive annuli surrounding each region and scaling it relative to the ROI extent. We then extracted the ROI-integrated spectra from the optimized PHANGS-MUSE NGC 628 'copt' data cube.

We removed the contribution of the underlying stellar population by fitting the spectra of each region with the Penalized Pixel-Fitting (pPXF) code (Cappellari 2023) over the wavelength range 4800 Å to 9250 Å, which includes the [S III]λ 9069 emission line. The pPXF fitting yields rectified spectra with the stellar continuum subtracted. We adopted a constant instrumental resolution of MUSE = 2.65 Å, despite its mild wavelength dependence (Bacon et al. 2017), and enforced a tolerance of five consecutive missing spectral values in the spectra. We employed Surface Brightness Fluctuations templates from the E-MILES database (Vazdekis et al. 2012, 2020), assuming a Chabrier (2003) initial mass function with slope of 1.30 and adopting Teramo/BaSTI stellar isochrones. Since the E-MILES templates have higher intrinsic spectral resolution than the MUSE data, they are convolved to match the observational resolution of our spectra. The templates library spans ages from 0.03 to 14 Gyr and metallicities from [Z/H] = −2.27 to +0.40. The fitting scheme relies solely on multiplicative Legendre polynomials to avoid modifying the continuum.

Emission line fluxes were fitted for each ROI using the Line Measuring library (LiMe) (Fernández et al. 2024), assuming pseudo-Voigt profiles and a minimum S/N of 4 for line detection. Fluxes were corrected for dust attenuation using the O'Donnell (1994) extinction law with R_v=3.1. We adopted a theoretical F(Hα)/F(Hβ) ratio of 2.86 corresponding to Case B recombination (Osterbrock & Ferland 2006). The deprojected galactocentric distance of each region was computed assuming a face-on orientation and an inclination of 8.9° (Table 1).

Although our algorithm returns the number of pixels defining each ROI, this quantity is not used to estimate the radius. Instead, region volumes are inferred assuming asymmetry from their two-dimensional projection. A minimum bounding ellipse is fitted to estimate the area. The third spatial axis is approximated as the geometric mean of the semi-major and semi-minor axes.

3.2. Definition of the SIGNALS nebular region

Within the SIGNALS survey in NGC 628, (Rousseau-Nepton et al. 2018) integrated fluxes are corrected for DIG emission and subsequently fitted with pseudo-Voigt profiles within the ZoI. However, while Hα fluxes are integrated out to the outer boundary of each region, emission line fluxes are limited to an aperture of radius $\sigma_{\rm PVoigt}$ to minimize DIG contamination and crowding effects. This can lead to an overestimation of Hα luminosity relative to emission lines fluxes, which could affect their ratios in diagnostic diagrams.

In the SIGNALS public catalog, each nebular region is modeled as circular, with the radius estimated by convolving the pseudo-Voigt σ with a Gaussian kernel to facilitate comparison with the classical spherical Strömgren model. To obtain radii consistent with the Hα integration limit in Sect. 3.1, we reconstructed the pseudo-Voigt $H\alpha$ intensity profile, assumed spherical symmetry, and adopted the radius of $\sqrt{2}\sigma_{Pvoigt}$, as a conservative size estimate. To take this into account in the derivation of the Hα flux of the H II regions, we computed a flux correction factor quantifying the fraction of the integrated Hα flux within the $\sqrt{2}\sigma_{Pvoigt}$ radius relative to the total within the outer boundary. The resulting Hα correction factors derived have a mean value of $\sim$0.75 $\pm$ 0.05. Equivalently, all selected radii exceed the instrumental resolution, taking the SIGNALS survey FWHM as the effective resolution limit.

Lastly, the galactocentric distances, R_g, are deprojected assuming a face-on orientation. While the public SIGNALS catalog adopts an inclination of 21°, we recalculated it for all regions using the same position angle and inclination as those adopted for the PHANGS-MUSE sample to ensure consistency.

3.3. BPT classification

We used the Baldwin–Phillips–Terlevich (BPT-type) diagnostic diagrams (Baldwin et al. 1981), together with the classification curves from Kewley et al. (2001) (Ke01) and Kauffmann et al. (2003) (Ka03), as an initial step to define our H II region subsample from the two surveys (Fig. G.1). This approach discriminates photoionized H II regions from nebulae dominated by alternative ionization mechanisms. Three diagnostic diagrams are commonly used: [N II]λ6583/Hα versus [O III]λ5007/Hβ (BPT-NII); [S II]λ6716 + λ6731/Hα versus [O III]λ5007/Hβ (BPT-SII); and [O I]λ6300/Hα versus [O III]λ5007/Hβ (BPT-OI). Since the SIGNALS survey does not include the [O I] λ6300 emission line, our analysis is restricted to the BPT-NII and BPT-SII diagrams. We applied conservative selection criteria following Kewley et al. (2006), retaining only regions located below and to the left of the Ka03 classification line in the BPT-NII diagram, and below and to the left of the Ke01 line in the BPT-SII diagram.

To minimize the impact of low signal-to-noise ratios, we required S/N > 4 for all relevant strong emission lines. The adopted criteria are summarized in Table F.1 of Appendix F.

In the SIGNALS survey, a small difference in the spatial integration of fluxes could remain after Hα flux correction, since the forbidden lines are measured within $\sigma_{\rm PVoigt}$ (Rousseau-Nepton et al. 2018) while the corrected Hα corresponds to $\sqrt{2}\,\sigma_{\rm PVoigt}$. Therefore, a slight underestimation of the [N II]/Hα and [S II]/Hα ratios may still be present.

The SIGNALS sample and the PHANGS sample we present (hereafter PHANGS) consist of H II regions with an $R^2_{fit} > 0.65$ to ensure a consistent comparison between them. This threshold discards regions with poorly constrained radial profiles in SIGNALS and ensures that the retained H II regions in both catalogs have comparably well-constrained radii and Hα luminosities.

3.3.1. Identification of supernova remnants

To ensure that all regions included in our sample are H II regions, we identified and removed potential supernova remnants (SNRs).

SNRs can contribute significantly to the Hα emission through shock-driven ionization. In spatially integrated Integral Field Unit (IFU) spectra, the coexistence of photoionized and shock-ionized gas, according to recent work (Points et al. 2019; Duarte Puertas et al. 2024) can complicate the identification of SNRs using BPT-type diagnostics alone. We therefore adopted a set of complementary selection criteria, following Moumen et al. (2019), to minimize contamination from shock-dominated sources.

Additionally, we examined the velocity dispersion of the Hα and [S II]λ6716 emission lines, correcting for instrumental broadening as detailed in Li et al. (2024). Shock-heated gas in SNRs produces broader emission line profiles than those typical of H II regions, which generally exhibit a velocity dispersion of 20 km s^{-1} (Winkler et al. 2017). This additional kinematic criterion is applied only to the PHANGS-MUSE sample, as velocity dispersion measurements are not available for the SIGNALS survey. For Hα and [S II]λ6716, the observed velocity dispersion was defined as an equivalent dispersion derived from the Gaussian FWHM profile ($\sigma = FWHM/(2\sqrt{2\ln 2})$. Instrumental broadening is removed in quadrature using the wavelength-dependent line-spread function from Bacon et al. (2017), yielding mean instrumental dispersions of $\sigma_{\rm inst}(H\alpha)$ = 48.9 km s^{-1} and $\sigma_{\rm inst}([S\,II]6716)$ = 48.7 km s^{-1}.

The adopted SNR identification criteria are listed below:

1. Emission line ratio [S II]λ6716 + λ6731/H$\alpha \geq 0.4$ (Mathewson & Clarke 1973).
2. Maximum physical size of $\sim$120 pc (Cajko et al. 2009).
3. Classification in Sabbadin diagnostic diagrams (Sabbadin et al. 1977), following Leonidaki et al. (2012)
4. Location outside the H II region domain in at least one BPT diagram.
5. [O III]λ5007/Hβ < 6 (Cox & Raymond 1985; Hartigan et al. 1987).
6. S/N([S II]λ6716) > 20 and $\sigma_{[S\,II]\lambda6716} > 50$ km s^{-1} (Li et al. 2024).

3.3.2. DIG regions and visual inspection

Previous studies have shown that forbidden line ratios such as [S II]λ6716/Hα and [N II]λ6584/Hα reach significantly higher values in the DIG than in classical H II regions (e.g., Madsen et al. 2006; Haffner et al. 2009; Della Bruna et al. 2020). DIG emission may also exhibit line ratios similar to those of SNRs (Elwert & Dettmar 2005). We therefore examined whether our regions display elevated ratios indicative of substantial DIG contamination, noting that low FF values are also characteristic of DIG structures, despite their typically low electron densities of $\sim$ 0.1 cm^{-3} (Ferrière 2001). We adopted conservative DIG-line thresholds of $Log([S\,II]\lambda6716/H\alpha) > -0.55$ (Belfiore et al. 2022) and $Log([N\,II]\lambda6583/H\beta) > -0.35$ (Flores-Fajardo et al. 2009), corresponding to values 0.5 dex below the canonical DIG limits.

We visually inspected all H II regions included in the PHANGS sample to remove cases in which the analytically defined region extended significantly into the DIG, despite being

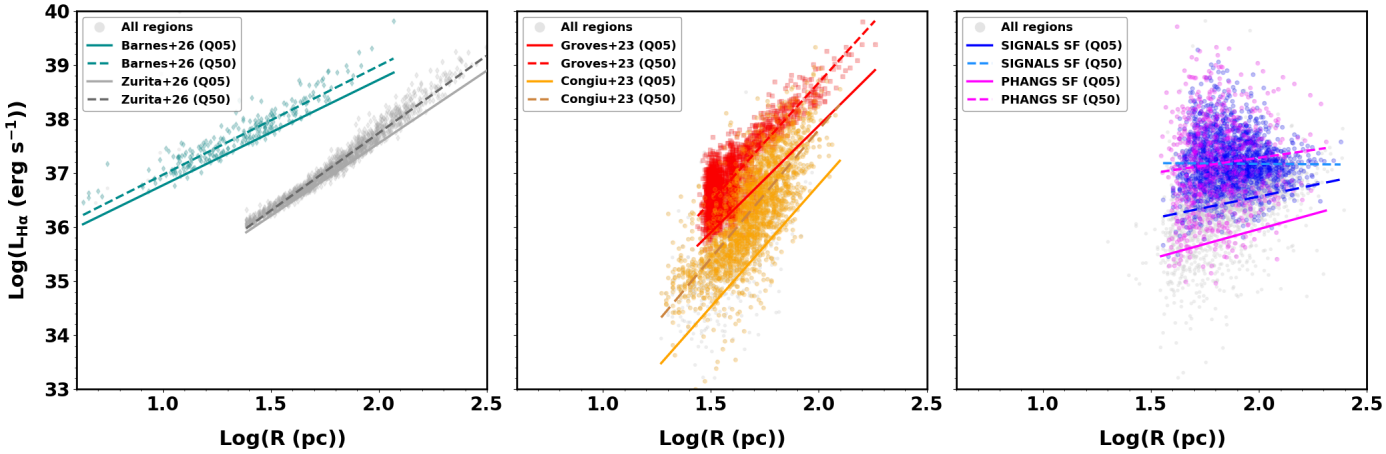


Fig. 1. Hα luminosity, $L_{H\alpha}$, against the nebula radius in NGC 628 for six published H II region catalogs. A homogeneous comparison via lower 5% (Q05, solid lines) and median (Q50, dashed lines) quantile regressions of $\log(L_{H\alpha})$ on $\log(R)$ is shown. Left panel (a): Barnes et al. (2026) (darkcyan diamonds, $\kappa_{05,50} \sim 1.95, 2.01$) and Zurita et al. (2026) (dark gray diamonds, uncorrected fluxes, $\kappa_{05,50} \sim 2.69, 2.87$), both with constrained absolute-threshold radius definitions; the Zurita et al. (2026) catalog already consists exclusively of H II regions and therefore has no gray background population. Middle panel (b): Groves et al. (2023) (red squares, not DIG-corrected, $\kappa_{05,50} \sim 3.95, 4.40$) and Congiu et al. (2023) (orange circles, DIG-corrected, $\kappa_{05,50} \sim 4.53, 4.76$), both with absolute-threshold radius definitions. Right panel (c): Rousseau-Nepton et al. (2018) (SIGNALS, blue dots, $\kappa_{05,50} \sim 0.83, -0.03$) and the PHANGS sample (magenta circles, $\kappa_{05,50} \sim 1.66, 0.96$), both with profile-truncated radius definitions ($R^2_{\rm fit} > 0.65$ for SIGNALS). In each panel, objects beyond the BPT H II region zone of the parent catalog are shown as light gray points (see Sects. 4 and A for details).

classified as H II regions based on their BPT line ratios. As a result, approximately 7% of the initial PHANGS-H II region sample was discarded, primarily due to very irregular *Hα* morphologies (e.g., crossing or filamentary structures) or excessively extended radii dominated by diffuse emission.

4. Luminosity–size coupling induced by the definition of the H II region

The Hα luminosity–size relation is commonly used to characterize the physical scaling of H II regions. Here, we compared the resulting luminosity–size relation for all considered catalogs of nebular regions, restricting each sample to star-forming (H II) regions identified using standard BPT-type diagnostics in order to minimize contamination from non-photoionized sources. The Hα luminosities used in this comparison have not been corrected for the observed dependence on galactocentric radius. The effect of this correction on the luminosity–size relation is examined separately in Appendix E. Fig. 1 presents the log($L_{H\alpha}$)-log(R) relation for H II regions in NGC 628 in the six catalogs we considered, arranged in three panels. For each catalog, we performed the corresponding lower 5% (Q05) and median (Q50) quantile-regressions. We denote the corresponding slopes as κ_{05} and κ_{50}[3], respectively. The fitted slopes and intercepts, with 1000 bootstrap uncertainties, are listed in Table 2. For comparisons among catalogs, we adopted κ_{05} as the reference estimator because it traces the lower envelope of the coherent H II region population at a given radius, while being less sensitive to the vertical scatter introduced by variations in emissivity among individual regions (Aragüete Riesco et al., in prep.).

The three panels show different behavior: the power-law slope κ of the log($L_{H\alpha}$)–R relation does not appear to be a universal physical quantity, but rather a methodological observable that encodes the degree of coupling between the adopted radius definition and the log($L_{H\alpha}$) of each region. The catalogs span nearly two orders of magnitude in their range of H II region radius, from the ∼10 pc resolution limit of Barnes et al. (2026) to the ∼170 pc scale of the MUSE-based catalogs. The most striking differences are found in the scatter of the log($L_{H\alpha}$)-R relation itself. In Rousseau-Nepton et al. (2018) and the PHANGS sample (panel c), the slopes are shallow and even compatible with zero considering κ_{50}, with $\kappa_{05} \sim 0.83$–1.66 at Q05 flattening to $\kappa_{50} \sim -0.03$–0.96, accompanied by a pronounced vertical scatter at fixed R and a triangular "Christmas-tree" distribution. On the other hand, Groves et al. (2023) and Congiu et al. (2023) (panel b) produce steep slopes of $\kappa_{05} \sim 3.95$–4.53 with a tighter, collapsed distribution. Barnes et al. (2026) and Zurita et al. (2026) (panel a) occupy intermediate positions with $\kappa_{05} \sim 1.95$–2.69.

This diversity does not seem to reflect intrinsic differences in the physical properties of nebular regions in the catalogs. This is illustrated by the fact that Congiu et al. (2023) and the present work construct their nebular catalogs from the same Hα frame but employ different pipelines for the adopted radius definition. These differences arise because each segmentation

Table 2. Quantile regression fits to the $\log L_{H\alpha}$–$\log R$ relation for star-forming regions in each catalog.

Catalog	Quantile	Slope (κ)	Intercept	N
Barnes+26	κ_{05}	1.95 ± 0.09	34.82 ± 0.14	245
Barnes+26	κ_{50}	2.01 ± 0.07	34.96 ± 0.09	245
Zurita+26	κ_{05}	2.69 ± 0.03	32.18 ± 0.06	1458
Zurita+26	κ_{50}	2.87 ± 0.02	32.00 ± 0.03	1458
Groves+23	κ_{05}	3.95 ± 0.35	29.98 ± 0.55	2459
Groves+23	κ_{50}	4.40 ± 0.06	29.87 ± 0.10	2459
Congiu+23	κ_{05}	4.53 ± 0.18	27.73 ± 0.29	2342
Congiu+23	κ_{50}	4.76 ± 0.09	28.29 ± 0.14	2342
SIGNALS	κ_{05}	0.83 ± 0.21	34.91 ± 0.39	1780
SIGNALS	κ_{50}	−0.03 ± 0.09	37.23 ± 0.18	1780
PHANGS	κ_{05}	1.66 ± 0.35	32.72 ± 0.63	761
PHANGS	κ_{50}	0.96 ± 0.29	35.41 ± 0.53	761

[3] κ_{50} is shown for comparison with standard fits.

algorithm defines a region radius using criteria that introduce different levels of coupling between the adopted radius and the underlying Hα surface brightness (Wisnioski et al. 2012; McClain et al. 2026), which can systematically tune the resulting luminosity–radius relation.

None of the methods discussed here should be intrinsically superior to the others from the standpoint of physical fidelity. Each approach makes well-defined, if different, operational choices about what constitutes the radius of a nebula, and each is internally self-consistent. As demonstrated analytically in Appendices B and C, the slope κ propagates the methodological contribution to the observed slopes of the log(FF)–log(R) and log(FF)–log($L_{H\alpha}$). Any departure of an observed slope from its predicted methodological log($L_{H\alpha}$)–R coupling value therefore constitutes a physical signal, independent of the catalog construction (Eq. (B.4) in Appendix A, and Eq. (C.3) in Appendix C).

We provide the basis for a direct, quantitative separation of physical and methodological effects in catalogs with fundamentally different radius definitions, predicting how the adopted radius definition propagates into derived relations such as log(FF)–log(R) and log(FF)–log($L_{H\alpha}$). We now describe the specific segmentation approach adopted by each catalog.

Rousseau-Nepton et al. (2018) and the PHANGS sample pipeline determine the integration radius based on the local flux–density diagram or radial profile structure, independent of the absolute flux level. In Rousseau-Nepton et al. (2018), the threshold is universal, set to $R_{\rm cut} = \sqrt{2}\sigma_{\rm Pvoigt}$ (Sect. 3.2), whereas in the PHANGS sample we adopted an adaptive, hierarchical approach combining profile fitting, morphology, and flux thresholds (Sect. 3.1), yielding $R_{\rm cut}$ values comparable to the previous method because it is also based on geometrical refinements applied to initial radii derived from $\sigma_{\rm Pvoigt}$. Accordingly, in the two profile-truncated methods, the region radius is only partially coupled to the total Hα luminosity, since the truncation is set relative to the local profile structure rather than to an absolute surface brightness threshold.

In contrast, Groves et al. (2023) identify nebular regions in PHANGS-MUSE data using a HIIphot-based pipeline (Thilker et al. 2000), which grows regions iteratively from local maxima until the surface brightness falls below an absolute emission measure threshold of EM_{min} = 0.5 pc cm^{-6}. The estimated radius is defined as the circularized equivalent of the segmented area, $R_{\rm cir} = \sqrt{\frac{A}{\pi}}$. Congiu et al. (2023) apply CLUMPFIND (Williams et al. 1994) to the same PHANGS-MUSE data, identifying regions through intensity contours down to a threshold of background plus noise dispersion, and then applying a probabilistic, machine-learning algorithm to refine the classification of detected nebulae and distinguish H II regions from other ionized sources. In both cases the radius of nebular regions is defined as the circularized equivalent of the segmented area, A, $R_{\rm cir} = \sqrt{A/\pi}$, and $R_{\rm cir}$ grows proportionally with the absolute flux level because the radius of the nebulae is determined by an absolute threshold applied to the same flux map that is integrated to obtain the Hα luminosity. This introduces a strong coupling between R and $L_{H\alpha}$.

Barnes et al. (2026) construct a catalog of H II regions in PHANGS galaxies by applying a top-down segmentation to the PHANGS-MUSE nebula catalog Groves et al. (2023) to HST Hα imaging homogenized to a physical resolution of 10 pc and a spatially constant noise level $\sigma_{\rm final,10pc}$. Then, source identification follows a two-threshold procedure: compact structures detected above a high threshold of $5\sigma_{\rm final}$ are grown down a lower threshold of $2\sigma_{\rm final}$. $\sigma_{\rm final}$ is spatially uniform by selection, then the effective detection threshold is absolute rather than local. However, the growth of each region is topologically confined within the parent MUSE nebula mask, which prevents the radii from expanding as in the pure and unconstrained absolute threshold methods of Barnes et al. (2026) and Congiu et al. (2023). This topological constraint partially decouples $R_{\rm cir}$ from the absolute flux level, yielding an intermediate $L_{H\alpha}$-R slope of $\kappa_{05} \sim 1.95$. Here, the radius is defined identically as $R_{\rm cir}$ from the segmented area. In the Barnes et al. (2026) catalog only 216 regions were ultimately retained and positively identified as H II regions. However, the dominant factor distinguishing this catalog from the others is its higher spatial resolution. At ~10 pc, HST Hα imaging resolves H II region substructure approximately four times more finely than the ~35–44 pc PSF-limited SIGNALS and PHANGS-MUSE data, yielding intrinsically more precise size measurements and, consequently, a tighter and systematically offset $L_{\rm H\alpha}$–R relation (see also Sect. 6.1). The additional selection criteria likely contribute to the higher luminosities of the retained regions, whereas the improved spatial resolution primarily explains the tighter relation and its offset with respect to the other catalogs.

Finally, Zurita et al. (2026) constructed H II region catalogs for NGC 628 using image segmentation based on the spatial gradient of the Hα surface brightness. A background map tracing the diffuse ionized gas (DIG) emission is created by masking pixels above a gradient cut-off. H II regions are identified as structures with emission exceeding $3\sigma_{\rm rms-noise}$ in the background-subtracted image, where $\sigma_{\rm rms-noise}$ is approximately uniform. The method belongs to the class of photometrically constrained absolute threshold techniques, partially decoupling $R_{\rm cir}$ from the absolute flux and limiting the expansion of the radii compared to pure absolute threshold methods. Radii are circularized, defined as $R_{\rm eq} = \sqrt{\rm A/\pi}$. This approach yields an intermediate Hα luminosity–radius slope of $\kappa_{05} \approx 2.69$ for NGC 628.

Based on the type of coupling between the adopted radius definition and the Hα luminosity, the six catalogs compared in this section fall into three distinct methodological categories, which we analyzed in Appendix A:

1. Profile-truncated methods (Rousseau-Nepton et al. (2018) and the PHANGS sample): the integration radius is defined by the local structure of the radial brightness profile, fixing $R_{\rm cut} \approx k\sigma$ independently of the absolute surface brightness Σ_0. The $L_{H\alpha}$-R slope reflects the intrinsic variation of the peak surface brightness with region size, yielding $\kappa_{05} \sim$ 0.83–1.66.
2. Absolute-threshold methods (Groves et al. (2023) and Congiu et al. 2023): the radius grows until the surface brightness falls below a fixed, universal threshold $\Sigma_{\rm min}$, introducing a strong coupling between $R_{\rm cir}$ and $L_{H\alpha}$ that depends on the ratio $\Sigma_0/\Sigma_{\rm min}$. This produces steep $L_{H\alpha}$–R slopes of $\kappa_{05} \sim$ 3.95–4.53.
3. Hybrid methods (Barnes et al. (2026) and Zurita et al. (2026)): the coupling between R and $L_{\rm H\alpha}$ is partially suppressed by a secondary constraint, yielding intermediate slopes $\kappa_{05} \sim$ 1.95–2.69. In Barnes et al. (2026) this suppression is topological: the region growth is confined within a parent MUSE mask, limiting the maximum extent independently of the flux level. In Zurita et al. (2026) the suppression is photometric: the detection threshold adapts to the background-subtracted image. Despite having similar κ_{05}, the two mechanisms are different and are expected to produce distinct behaviors.

The three families of methods yield systematically different power-law log($L_{H\alpha}$)–R slopes, which reflect the extent to which the adopted radius definition couples the measured radius to the intrinsic flux distribution of each region. This is derived analytically in Appendix A. Any deviation of an observed slope from its predicted methodological value therefore constitutes a physical measurement, independent of the radius definition. This interpretation should, however, be considered in the context of additional methodological differences among catalogs beyond the adopted radius definition, such as the application of DIG corrections (not applied in the case of the Groves et al. (2023) catalog) and differences in the post-processing pipelines used to derive integrated fluxes.

We next examine the impact of different radius definitions on the log($L_{H\alpha}$)–log(R) distribution. The vertical scatter in the log($L_{H\alpha}$)–log(R) plane at fixed radius (the "Christmas-tree" distribution) visible in the left panel of Fig. 1 illustrates that H II regions of a given Hα luminosity can span a broad range of radii, and conversely, regions of similar size can exhibit a wide range of Hα luminosities.

Eq. (7) shows that at fixed R, the luminosity varies because FF and n_e both differ in the population. This diversity would be preserved only when the radius definition does not couple R to $L_{H\alpha}$. Profile-truncated methods (Rousseau-Nepton et al. (2018) and this work) determine the integration radius from the local radial brightness profile. Consequently, R does not systematically increase with $L_{H\alpha}$, allowing the full range of FF and n_e to appear as vertical scatter. The triangular shape arises because the lowest luminosities measured are limited by the detection limit (PSF). The observed upper-right limit implies that any region above this limit would be a very large, highly luminous region and would require an Hα luminosity so extreme that either no coherent region can exist or stellar feedback disperses the gas before it can be observed.

By contrast, absolute surface brightness thresholds (Groves et al. (2023) and Congiu et al. (2023)) assign radii that grow with peak flux. Brighter regions have larger $R_{\rm cir}$, stretching the scatter. Here, faint compact and luminous extended regions cannot coexist at the same radius, collapsing the "Christmas-tree" into a narrow, nearly one-dimensional distribution.

An intermediate case is found in Zurita et al. (2026) ($\kappa \approx 2.69$). When κ approaches 3, it implies that the assigned radius for each region scales approximately as $R \propto L_{H\alpha}^{1/3}$. In practice, faint regions are assigned small radii while luminous regions receive large radii. The radius definition thus imposes a tight functional link between $L_{H\alpha}$ and R, leaving little room for vertical scatter at fixed R. The Christmas-tree shape collapses because the method prevents compact regions from being highly luminous and extended regions from being faint, by assigning radii that closely track the intrinsic surface brightness of each region, compressing scatter perpendicular to the $L_{H\alpha}$–R relation.

The analysis from this point onwards is restricted to five catalogs, as the [S II] emission line fluxes in Zurita et al. (2026) are not provided for NGC 628.

5. Physical parameters and sample definition

In this section, we describe how we determined the parameters involved in the FF calculation, together with the physical properties of the H II regions. We first estimate the local electron density, which is an important property of H II regions and provides key information for characterizing star-forming regions and studying the spatial variation of the interstellar medium (ISM) within the host galaxy. We derived the mean luminosity-weighted electron density $\langle n_e \rangle$ and selection criteria required for the FF analysis, including the definition of the characteristic radius and the final sample selection. Finally, we then compare the local electron density, luminosity and size distributions of H II regions across the five catalogs.

5.1. Local electron density

To compare H II regions based on their intrinsic physical characteristics, we first estimate their local, or *insitu*, electron density using the ratio of the collisionally excited [S II] emission lines at optical wavelengths: [S II] λ6716 and [S II] λ6731.

For the SIGNALS and PHANGS samples we estimated the local electron density (n_e) from the intensity ratio of the [S II] emission lines $\lambda\lambda 6716, 6731$, using the analytical solution for a three-level atom (McCall 1984). This ratio is related to the electron density through

$$\mathrm{R_{[S_{II}]}} \sim 1.49 \left[\frac{1 + 3.77\mathrm{x}}{1 + 12.8\mathrm{x}} \right] \tag{11}$$

where $\mathrm{x} = 10^{-4} \mathrm{n_e t^{-1/2}}$ with $\mathrm{t} = \mathrm{T_e}/10\,000$. Assuming a typical gas temperature in the nebula of $\mathrm{T_e} \sim 10^4$ K for star-forming regions (thus t = 1), the expression simplifies to

$$\mathrm{n_e} = \left[\frac{1.49 - \mathrm{R_{[S_{II}]}}}{12.8\mathrm{R_{[S_{II}]}} - 5.6173} \right] \times 10^4. \tag{12}$$

This method is valid over the density range $10 - 10^4$ cm^{-3} (Castaneda et al. 1992; Zaritsky et al. 1994).

5.2. rms electron density

We calculated the mean luminosity-weighted *rms* electron density, $\langle n_e \rangle$, for each region from the $H\alpha$ emission line flux. In PHANGS-H II, the radius R was calculated as the geometric mean of the semi-major and semi-minor axes of the minimum bounding ellipse for each region, which corresponds to the radius of a circle with the same area. We assumed a constant electron temperature of 10^4 K for the H II regions (Osterbrock 1989; Gutiérrez & Beckman 2010),

$$\langle n_e \rangle = 1.5 \times 10^{-16} \left[\frac{L_{H\alpha}}{R^3} \right]^{1/2} \quad \mathrm{cm^{-3}}. \tag{13}$$

The $\langle n_e \rangle$ measurements show no significant variation with galactocentric radius, suggesting that the luminosity-weighted electron density is largely insensitive to the large-scale galactic environment before correcting for the radial luminosity gradient.

Direct cross-catalog comparisons of the $\langle n_e \rangle_{\rm rms}$–$R$ slope are not straightforward because the observed gradient depends on the $L_{H\alpha}$–R coupling parameter κ, which varies with radius definition.

5.3. Final H II region samples

We defined the final H II region samples of SIGNALS and PHANGS by applying a set of quality and physical criteria. We adopted a conservative lower limit of $n_e \sim 80$ cm^{-3} to ensure reliable electron density estimates. In addition, we required a minimum signal-to-noise ratio in the emission lines used for the derivation of physical parameters, as summarized in Table F.1.

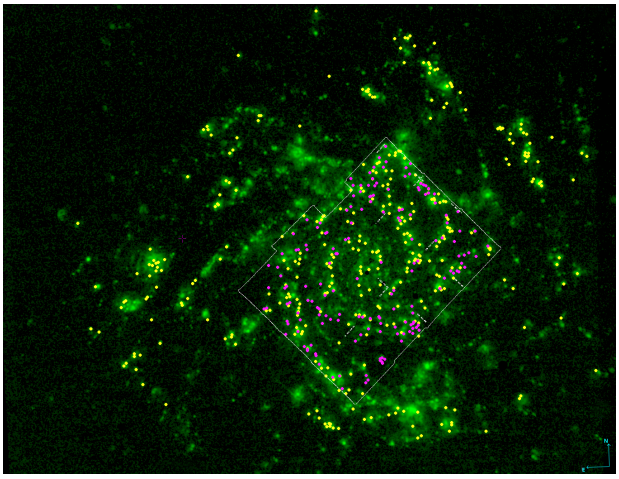


Fig. 2. SIGNALS Hα image of NGC 628 showing the final selection of H II regions from SIGNALS-H II (yellow dots) and from PHANGS-H II (magenta dots). The white contour delineates the PHANGS-MUSE mosaicked field of view. North is up, and east is to the left.

For the FF calculation, the characteristic radius R is defined according to the catalog method. In SIGNALS, R corresponds to the profile-truncated radius derived from the pseudo-Voigt fit, while in PHANGS it is defined as the geometric mean of the semi-major and semi-minor axes of the minimum bounding ellipse, yielding an area-equivalent circular radius. The filling factor is then computed following Eq. (6). We further excluded regions with log(FF) < -7, which are likely dominated by observational uncertainties. The resulting samples consist of 475 H II regions in SIGNALS-H II and 147 in PHANGS-H II.

To assess the consistency between electron density measurements in the two surveys, we constructed a new subsample (hereafter SIGNALS-MUSE) containing 227 regions from SIGNALS-H II sample. We created this subsample by limiting it to regions fully enclosed within the PHANGS-MUSE FoV, maintaining a minimum distance of 1 PSF from the mosaicked boundaries. Inspection of the right histogram in Figure G.2 (see Appendix) shows that both subsamples show a concentration of values near the adopted lower density limit 80 cm^{-3}. Fig. 2 shows the Hα image of NGC 628 with the final selection of H II regions from SIGNALS-MUSE and PHANGS-H II. To enable a direct comparison of the electron densities in the catalogs, we derived n_e for the samples of Groves et al. (2023), Congiu et al. (2023), and Barnes et al. (2026), as described in Subsubsect. 5.4.1. We applied a common set of quality criteria following the same procedure used for the SIGNALS-H II and PHANGS-H II catalogs (i) classification as H II according to BPT-type diagnostics; (ii) positive fluxes in the two [S II] $\lambda\lambda$6716, 6731 lines; (iii) no edge or stellar-contamination flags when available; (iv) signal-to-noise ratios greater than four in [S II] λ6716, [S II] λ6730, [N II] λ6583, [O III] λ5007, Hα, and Hβ; and (v) H II region radii larger than the PSF FWHM. Furthermore, for the catalogs of Congiu et al. (2023), Groves et al. (2023), and Barnes et al. (2026), the FF calculation assumes that R corresponds to the circularized radius derived from the segmented area.

These criteria reduced the parent samples to 391 regions out of 2855 in Groves et al. (2023), 704 out of 3502 in Congiu et al. (2023), and 216 out of 258 in Barnes et al. (2026). The FF analysis was further restricted to regions with electron densities based on the McCall (1984), $n_e \geq 80$ cm^{-3}, yielding final samples of 27, 192, and 29 regions for Groves et al. (2023), Congiu et al. (2023), and Barnes et al. (2026), respectively.

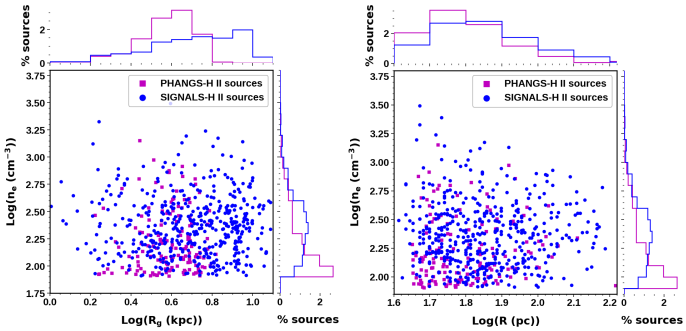


Fig. 3. In situ electron density, n_e, against the galactocentric radius, R_g of NGC 628 (left panel) and against the size of the region estimated through the profile-truncated radii, R, (right panel) for Rousseau-Nepton et al. (2018) (SIGNALS, blue dots) and this work (magenta squares). The radii are defined by a criterion relative to the radial profile structure in the flux-distance diagram. In Rousseau-Nepton et al. (2018), R corresponds to a universal $\sqrt{2}\sigma_{Pvoigt}$ for all the segmented regions, and in this work, R emerges from the pseudo-Voigt adjustment and the hierarchical selection of the integration radius based on geometric and dispersion criteria. Regions below the low-density limit ($n_e < 80$ cm^{-3}) are not shown.

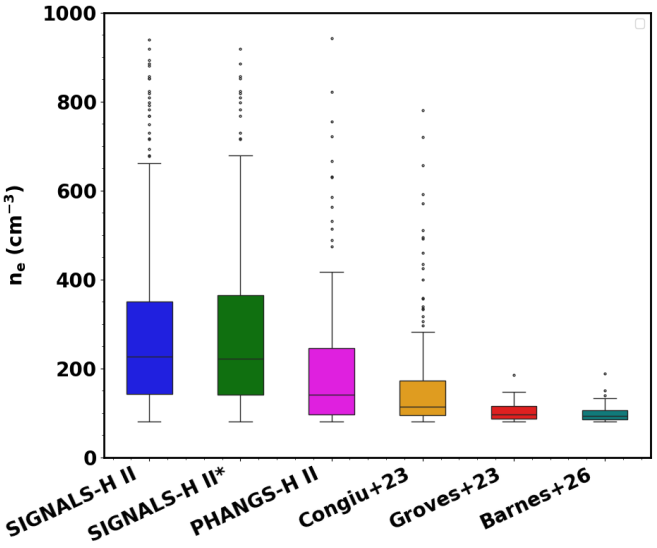


Fig. 4. n_e boxplot distributions for H II regions in NGC 628 in five catalogs: SIGNALS-H II, SIGNALS-H II* (where the asterisk indicates the subsample limited to sources located within the MUSE-mosaicked FoV), PHANGS-H II, Groves et al. (2023); Congiu et al. (2023); Barnes et al. (2026). The electron density is derived from the [S II] $\lambda\lambda$ 6716/6731 doublet ratio following the analytical solution of McCall (1984), assuming $T_e = 10^4$ K in all catalogs.

5.4. Comparison of the H II region catalogs

5.4.1. Local electron density

The left panel of Fig. 3 shows that the local electron density exhibits no significant dependence on galactocentric radius. In the right panel of the same figure, the PHANGS-H II regions appear somewhat more populated at H II smaller radii. This trend likely reflects improved detection techniques within the mosaicked MUSE field, which enables the detection of a larger number of compact regions at small galactocentric radii.

Fig. 4 shows the n_e distributions for the five catalogs. Groves et al. (2023) and Barnes et al. (2026) show the lowest

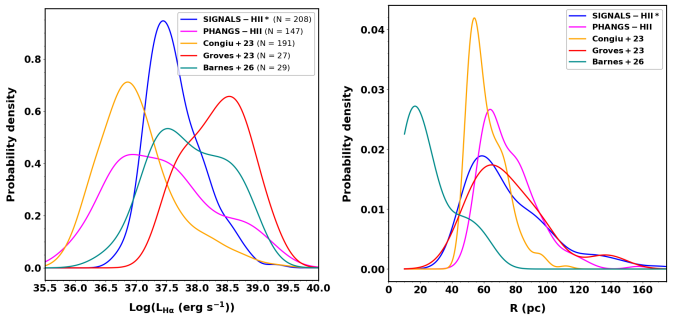


Fig. 5. Left panel: KDE of the Hα luminosity distribution, log(L$_{H\alpha}$), for H II regions in NGC 628 in all five catalogs: SIGNALS-H II* (where the asterisk denotes the subsample restricted to sources within the MUSE mosaic footprint), PHANGS-H II, and the catalogs of Congiu et al. (2023); Groves et al. (2023); Barnes et al. (2026). Right panel: distributions of the H II region radii in the same catalogs. The radius R corresponds to the profile-truncated radius for SIGNALS-H II. For PHANGS-H II, R is computed as the geometric mean of the semi-major and semi-minor axes of the minimum bounding ellipse fitted to each region, providing an equivalent circular radius. For the catalogs of Congiu et al. (2023), Groves et al. (2023), and Barnes et al. (2026), the radii correspond to the equivalent circular radius derived from the number of pixels within the segmented area. In Barnes et al. (2026), below 10 pc, the radius definition is dominated by resolution effects.

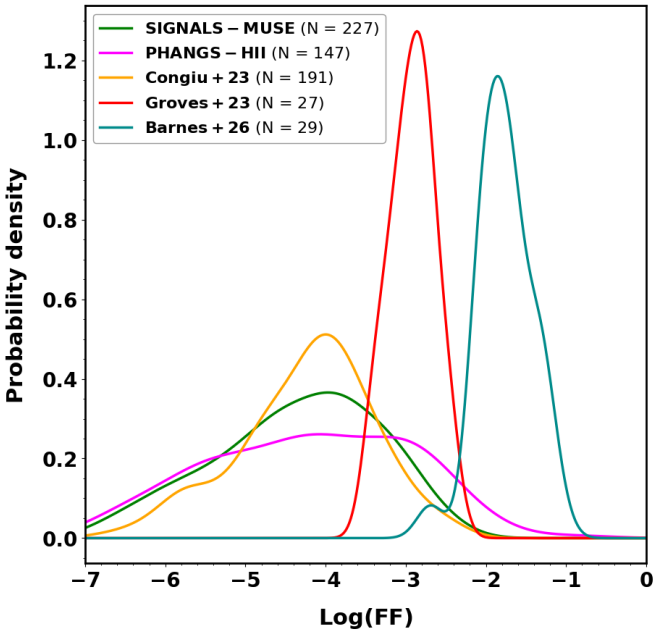


Fig. 6. KDE of the log(FF) distribution of H II regions in NGC 628 among all the different catalogs: SIGNALS-H II, PHANGS-H II, Groves et al. (2023); Congiu et al. (2023); Barnes et al. (2026). The overall range of FF values is comparable in the PHANGS-MUSE and SIGNALS catalogs, although the median values shift systematically depending on the adopted boundary definition and spatial resolution.

median densities (96.8 ± 11.3 and 93.8 ± 10.6 cm^{-3} respectively, with uncertainties given by the median absolute deviation), Congiu et al. (2023) and PHANGS-H II show intermediate median comparable densities (113.8 ± 27.2 and 140.84 ± 50.1 cm^{-3} respectively), consistent with samples spanning a wider range of physical conditions. The SIGNALS-H II catalog shows the highest median densities (231.1 ± 97.5 cm^{-3}). When the comparison is restricted to the spatial MUSE FoV, the median yields (211.9 ± 88.4 cm^{-3}; see also Fig. G.2).

The observed offset should not be interpreted solely as evidence of intrinsic differences between the H II regions sampled. Instead, it likely results from a combination of observational and methodological effects, including differences in spatial resolution, sensitivity, DIG treatment, flux integration, and region segmentation. At the same time, the PHANGS and SIGNALS catalogs are not fully overlapping samples owing to their different FoV and selection functions (Sect. 2).

This interpretation is consistent with the electron density distributions shown in Fig. G.2.

5.4.2. Luminosity and size distributions in the catalogs

We compared the kernel density estimate (KDE) Hα luminosity distribution in all catalogs. Fig. 5 illustrates the differences in luminosity and size distributions in all catalogs. In the left panel, the median Hα luminosities are ∼37.6, 38.3, and 37.9 erg s^{-1} for SIGNALS-H II restricted to the MUSE mosaicked FoV, Groves et al. (2023); Barnes et al. (2026), respectively, while Congiu et al. (2023) and PHANGS-H II reach fainter populations with medians of 36.9 and 37.4 erg s^{-1}. Considering the luminosity histograms in Fig. G.3 for SIGNALS-H II and PHANGS-H II, the SIGNALS-H II sources populate a similar luminosity range than PHANGS-H II but the latter recovers a larger number of intermediate- to low-luminosity regions.

Conversely, the DIG-uncorrected catalog of Groves et al. (2023) integrates fluxes out to nearly the full radial extent of the regions, which expand for brighter regions and increases the luminosity measurement, whereas PHANGS-H II truncates the integration at a profile-defined radius, which enables the detection of compact faint regions. The size distributions in the right panel of Fig. 5 show similar behavior. The radii histogram for SIGNALS-H II and PHANGS-H II catalogs in Fig. G.4 shows in the left panel that nearly 60% of SIGNALS-H II sources have radii smaller than 80 pc, with a long tail toward larger sizes. The PHANGS-H II subsample reveals that nearly 80% of regions have radii below 80 pc, with a notably higher fraction of small resolved regions at 50 pc.

Barnes et al. (2026) yields the smallest mean radius (26 pc), consistent with the HST-resolved substructures at ∼10 pc resolution. SIGNALS-H II and Groves et al. (2023) reach the largest mean radii (77.9 and 76.1 pc), reflecting boundaries that extend to large fractions of the Gaussian profile in bright regions. PHANGS-H II and Congiu et al. (2023) show intermediate values (60.3 and 62.1 pc), with Congiu et al. (2023) displaying the smallest dispersion (11.7 pc) among the MUSE-based catalogs.

6. Analysis and results

In this section, we present the FF values of H II regions in NGC 628 and investigate their relation with physical quantities. We analyze how FF varies with the radius of H II regions, local electron density, and Hα luminosity. We further compare its distribution in the different H II region catalogs and assess the range of physical conditions covered by the classical FF formulation. Finally, we investigate possible correlations with other structural properties, including a possible connection with PAH-to-dust emission.

6.1. Filling factor in H II regions

The range of FF values in Fig. 6 exhibits consistency among the SIGNALS-H II, PHANGS-H II and Congiu et al. (2023)

catalogs (The specific histogram comparing the FF distributions of SIGNALS-H II and PHANGS-H II is presented in Fig. G.5). However, the median values shift systematically depending on the adopted boundary definition and spatial resolution in SIGNALS-H II, PHANGS-H II, and the catalogs of Congiu et al. (2023); Groves et al. (2023); Barnes et al. (2026). Barnes et al. (2026) shows the highest median value, log(FF) = −1.8, followed by Groves et al. (2023) with −2.9, while SIGNALS-H II, PHANGS-H II and Congiu et al. (2023) go down to a lower median value of log(FF) ∼ −4.4, –4.2 and –4.1 respectively, with PHANGS-H II having the largest dispersion (σ = 1.4 dex). This ordering follows the differences in the characteristic radii discussed above (see Sect. 4). Barnes et al. (2026), which reports the smallest median radius, yields the highest FF values, whereas catalogs adopting larger boundary radii shift the distribution toward lower FF. There is substantial overlap between the SIGNALS-H II, PHANGS-H II, and Congiu et al. (2023) distributions.

The Groves et al. (2023) catalog does not include an explicit DIG subtraction, which may lead to systematically higher FF estimates, since diffuse emission within the aperture increases the measured Hα luminosity. In addition, DIG contamination may affect the [S II] emission line fluxes, which can bias the inferred electron densities toward lower values. These combined effects may therefore contribute to higher inferred FF values relative to the other catalogs.

To assess whether the systematically higher FF values reported by Barnes et al. (2026) reflect an extension of the same H II regions identified in PHANGS-MUSE catalogs toward smaller physical scales or are partly driven by the higher angular resolution of HST relative to MUSE/SITELLE, we estimate the impact of beam smearing on the inferred radii and derived FF values. Under the assumption of Gaussian profiles, the observed size of a region with intrinsic (HST-resolved) radius $R_{\rm true}$ can be approximated as the quadratic combination of the intrinsic size and the PSF full width at half maximum, Φ:

$$R_{\rm obs} \approx \frac{1}{2}\sqrt{(2R_{\rm true})^2 + \Phi^2}, \tag{14}$$

Adopting $\Phi \approx 35$ pc for SIGNALS/SITELLE and $\Phi \approx 43.78$ pc for PHANGS–MUSE (Sect. 2), and considering a region in Barnes et al. (2026) with a mean radius in NGC 628 of ∼26 pc, Eq. (14) yields $R_{\rm obs} \approx 34$ pc, corresponding to $\log(R_{\rm obs}/R_{\rm true}) \approx 0.12$ dex. Since FF $\propto R^{-3}$ at fixed $L_{\rm H\alpha}$ and n_e, this implies $\Delta\log({\rm FF}) \approx -3\log(R_{\rm obs}/R_{\rm true}) \approx -0.35$ dex. The observed offset between the median log(FF) measured by Barnes et al. (2026) and those derived for SIGNALS-H II, PHANGS-H II, and Congiu et al. (2023), amounting to ∼2.1–2.3 dex, can therefore be partially attributed to beam smearing. However, this effect accounts for a somewhat small fraction of the offset for sources near the population mean, increasing for the most compact regions close to the ∼10 pc HST resolution limit. The remaining discrepancy likely reflects the inclusion of high-surface-brightness H II regions that are resolved by HST but only partially resolved, or unresolved, in MUSE/SITELLE observations. We therefore interpret the elevated FF values reported by Barnes et al. (2026) as arising from a combination of resolution effects and a shift in the sampled FF distribution toward smaller and denser H II regions, without requiring a fundamentally distinct FF regime.

Beam smearing affects all seeing-limited datasets we considered, namely the SIGNALS and PHANGS-MUSE catalogs, although its impact is most pronounced when comparing these samples to the higher-resolution HST-based catalog of Barnes et al. (2026).

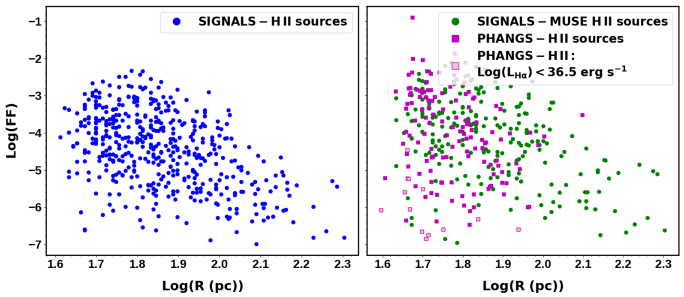


Fig. 7. FF against the radii of H II regions in log-log scale. We show the SIGNALS-H II regions in the left panel, and PHANGS-H II regions (magenta squares) along with SIGNALS-MUSE H II regions (green dots) in the right panel. The light magenta squares represent PHANGS-H II fainter regions with log(L$_{\rm H}\alpha$) < 36.5 erg s^{-1}. These fainter regions exhibit the greatest deviation from the trend.

6.2. Filling factor relations

The definitional scalings presented in Eq. (7) show that, by construction, FF is not independent of $L_{\rm H\alpha}$, R, and n_e. However, the observed trends may still carry physical information beyond this definitional scaling through the scatter, and we can use these relations as diagnostics of the regime in which the classical FF formulation provides a physically meaningful description. It is also necessary to account for the artificial trend introduced by the $Log(L_{H\alpha})$–$Log(R)$ coupling, which is addressed in Sect. 4.

Within this framework, we performed linear regression analyses to investigate correlations with L, R, and n_e. The fitted slopes and intercepts for the log(FF)–log(R^3), log(FF) – log(n$_e$), and log(FF)–log(L$_{\rm H\alpha}$) relations are summarized in Table 3 for the SIGNALS-H II and PHANGS-H II samples. The associated plots are Figs. 7, 10, 12.

6.2.1. Relation with the volume

Fig. 7 presents the correlation between FF and radii of H II regions in logarithmic scale. The low coefficients of determination obtained for the FF-volume relation ($R^2_{fit} \sim 0.07$–0.21; Table 3) indicate that this variable explains only a small fraction of the variance in FF. Although the definitional scaling is recovered in the slope values, the volume alone does not capture the intrinsic scatter of FF within the sample, which reflects variations in the L$_{\rm H\alpha}$/n$_{\rm e}^2$ and can provide insight into secondary trends among the other parameters. This behavior is consistent with Fig. 5, where for a given radius (or equivalently volume) the regions display a vertical dispersion reflecting variations in $FF\, n_e^2\, j_{H\alpha}(T_e)$ (Eq. (5)). The right panel of Fig. 8 shows the relation between log(FF) and radius, color-coded by n_e (left panel) and log(L$_{\rm H\alpha}$) (right panel). The color gradients confirm that the vertical scatter at fixed radius is physically meaningful: at a given radius, regions with higher n_e tend to exhibit lower FF, whereas more luminous H II regions cluster toward higher FF values. This suggests that regions with higher Hα luminosity tend to occupy the upper envelope of the FF distribution, but this relation is modulated by variations in electron density and region size. At the same time, lower-luminosity regions display a larger scatter around the classical FF relation.

Table 3. Linear regression models for the FF parameters.

Catalog	Model	Parameter	Values	Std. Error	t-value	p-value	Model Statistics
SIGNALS-H II	$\log(FF) = \alpha \log(n_e) + \beta$	β	1.84	0.25	7.38	<0.001	R^2_{fit} = 0.58, F-stat.=646.9
		α	−2.65	0.10	−27.51	<0.001	
	$\log(FF) = \gamma \log(R^3) + \delta$	δ	1.86	0.56	3.34	0.001	R^2_{fit}= 0.21, F-stat.= 130.2
		γ	−1.13	0.10	−11.41	<0.001	
	$\log(FF) = \eta \log(L_{H\alpha}) + \zeta$	ζ	−63.35	2.27	−28.41	<0.001	R^2_{fit} = 0.60, F-stat.=699.1
		η	1.60	0.06	26.44	<0.001	
PHANGS-H II	$\log(FF) = \alpha \log(n_e) + \beta$	β	2.48	0.84	2.97	0.004	R^2_{fit} = 0.42, F-stat.=46.64
		α	−2.72	0.40	−6.83	<0.001	
	$\log(FF) = \gamma \log(R^3) + \delta$	δ	2.24	1.96	1.15	0.25	R^2_{fit}= 0.07, F-stat.= 10.36
		γ	−1.18	0.37	−3.22	<0.001	
	$\log(FF) = \eta \log(L_{H\alpha}) + \zeta$	ζ	−49.89	3.07	−16.29	<0.001	R^2_{fit} = 0.63, F-stat.=224.0
		η	1.22	0.08	14.97	<0.001	

Notes. Luminosities are $H\alpha$ in erg s^{-1}, electron densities in cm^{-3}, volumes in pc^3; only PHANGS-H II regions with $Log(L_{H\alpha}) > 36.5$ erg s^{-1} are included in the fits. The Coefficient (Column 4) corresponds to the best-fit regression coefficient, with Std. Error (Column 5) indicating its 1σ uncertainty. The t-value (Column 6) is the ratio between the coefficient and its standard error, used to test whether the parameter is significantly different from zero, and the associated p-value (Column 7) gives the corresponding significance level. The coefficient of determination (R^2_{fit}; column 8) quantifies the fraction of variance explained by the model, while the F-statistic (Column 9) assesses the overall significance of the linear regression. R denotes the profile-truncated radius for SIGNALS-H II, and the geometric mean of the semi-major and semi-minor axes of the minimum bounding ellipse fitted to each region for PHANGS-H II.

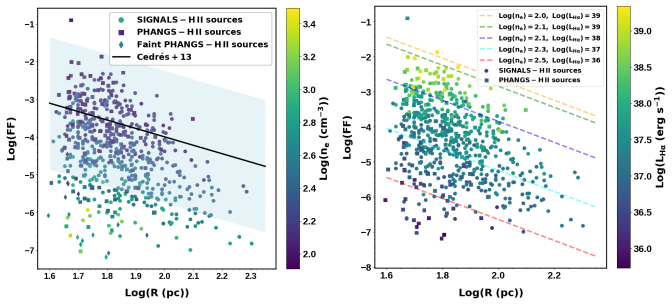


Fig. 8. Log(FF) as a function of the circularized equivalent radius, R, for SIGNALS-H II and PHANGS-H II regions in NGC 628, color-coded by the local electron density log(n_e) (left panel) and by the log($L_{H\alpha}$) luminosity (right panel). The trend reported by Cedrés et al. (2013) for H II regions in NGC 6946 is overplotted in the left panel with its 1σ dispersion for comparison at similar physical scales. Faint PHANGS-H II regions with log($L_{H\alpha}$) < 36.5 erg s^{-1} are highlighted with diamonds and exhibit the largest deviations from the trend, indicating a H II regime that is preferentially powered by single massive stars. (Sect. 6.2.1). The dashed lines in the right panel show the theoretical log(FF)–log(R) for selected log($L_{H\alpha}$) and log(n_e) values, assuming the Strömgren sphere model.

To quantify the physical contribution to the observed log(FF)–log(R^3) relation beyond the methodological effect introduced by the $L_{H\alpha}$–R coupling, we consider the observed slope as $\gamma^{obs} = \gamma^{pred} + \gamma^{phys}$, where $\gamma^{pred} = (\kappa_{05} - 3)/3$ represents the true physical slope expected (Appendix B) and κ_{05} is the slope of the $L_{H\alpha}$-R relation obtained from the 5th-quantile regression. For SIGNALS-H II, we obtain $\gamma^{obs} = -1.13$ and $\gamma^{pred} = -0.72$, which yield a physical residual of $\gamma^{phys} = -0.41$. For PHANGS-H II, the observed value is $\gamma^{obs} = -1.18$, with $\gamma^{pred} = -0.45$, and $\gamma^{phys} = -0.73$. The sign agreement between the two surveys is remarkable given their different instruments and spatial resolutions. This indicates that larger H II regions intrinsically exhibit lower FF, beyond the effect imposed by the radius definition. This behavior is consistent with an evolutionary scenario in which more extended regions correspond to later stages of expansion and dispersal of the ionized gas (Sect. 7).

The trend reported by Cedrés et al. (2013) for NGC 6946 is also shown in the left panel of Fig. 8 and the value reported is consistent with the more luminous H II regions. Low-luminosity PHANGS-H II regions (log($L_{H\alpha}$) < 36.5 erg s^{-1}, diamonds) systematically deviate below the Cedrés et al. (2013) relation and the theoretical Strömgren scaling. We calculated the average luminosity of these fainter regions in solar luminosity units[4], and inferred that the ionizing sources were consistent with single late O-type or early B-type stars (B0.5 or earlier, Vacca et al. (1996)). This behavior was further examined in the log(FF)–log(n_e) and log(FF)–log($L_{H\alpha}$) relations below, and departures in the three scaling relations were analyzed in the context of the classical FF framework in Sect. 7.

The apparent deviation of low-luminosity H II regions from the expected scaling does not necessarily imply a breakdown of the classical FF framework. Instead, it likely reflects observational sampling effects, including the limited spatial coverage of the MUSE observations for this galaxy. In other systems observed with the same pipeline, such as NGC 5068, a significantly larger population of faint and compact H II regions is recovered, showing that the classical FF formulation remains valid in this luminosity regime (Aragüete Riesco et al., in prep).

More generally, the observed trend may indicate a transition between different ionizing-source regimes. Depending on the distance of the galaxy, the detected H II regions preferentially sample nebulae powered by clusters or stellar associations rather than those ionized by single massive stars.

Fig. 9 compares the log(FF) distribution as a function of H II region radius for the five catalogs analyzed in NGC 628. The SIGNALS-H II, PHANGS-H II, and the catalogs of

[4] $L_\odot = 3.826 \times 10^{33}$ erg s^{-1}.

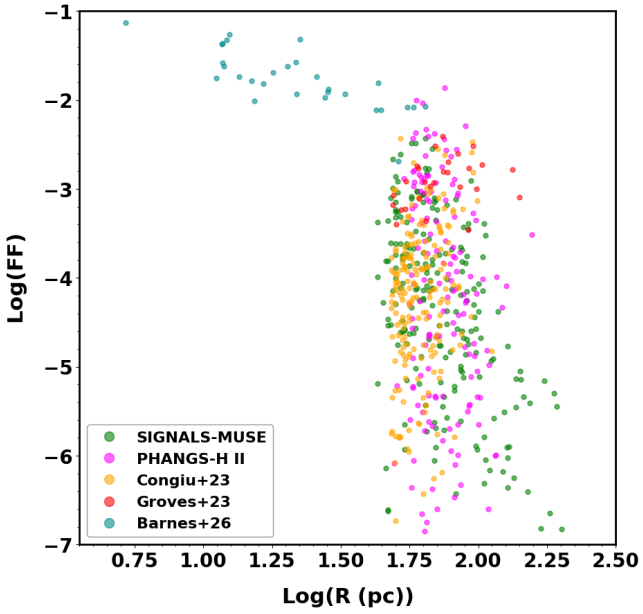


Fig. 9. Distributions of log(FF)–log(R) in NGC 628 among five catalogs: SIGNALS-MUSE, PHANGS-H II, Congiu et al. (2023); Groves et al. (2023) and Barnes et al. (2026).

Congiu et al. (2023) and Groves et al. (2023) span a similar range in radius, from approximately log(R) ∼ 1.6 to 2.1, and show a large dispersion in FF values. In contrast, the catalog of Barnes et al. (2026) probes smaller regions and shows comparatively higher FF values, likely reflecting resolution-driven effects. This suggests that differences among the seeing-limited catalogs are primarily methodological, while the HST-based catalog extends the analysis toward more compact and better-resolved H II regions.

Figure C.1 (see Appendix C) shows the log(FF)–log(R^3) relation for the four catalogs at comparable spatial resolution. Catalogs using truncated-profile radii (SIGNALS-H II and PHANGS-H II) show a consistent negative trend with substantial scatter, whereas absolute-threshold catalogs (Congiu et al. 2023, Groves et al. 2023) show flatter relations, with small scatter in Groves et al. (2023) and no clear trend in Congiu et al. (2023).

Analytically (see Appendix B), the expected slope of the log(FF)–log(R^3) relation is given by $\gamma = (\kappa_{05} - 3)/3$. After correcting for this methodological contribution, the residual slopes are $\gamma^{\rm phys} = -0.41$ for SIGNALS-H II and $\gamma^{\rm phys} = -0.73$ for PHANGS-H II, both indicating a decreasing trend of FF with region size. The difference between the two profile-truncated catalogs reflects the larger fraction of compact low-luminosity regions in PHANGS-H II, indicating that larger H II regions in NGC 628 tend to have intrinsically lower FF beyond what is imposed by the radius definition in all catalogs.

To investigate the apparent discrepancy in Congiu et al. (2023) ($\gamma^{\rm phys} = +0.26$), we repeat the analysis in bins of electron density ($n_e < 100$, $100 \leq n_e < 140$, and $n_e \geq 140$ cm^{-3}). In all bins, Congiu et al. (2023) yields $\gamma^{\rm phys} < 0$ (−0.33, −0.39, 0.05), consistent with SIGNALS-H II in the lower-density regimes. The positive value in the highest-density bin is driven by the limited dynamic range at large n_e (Fig. 4). PHANGS-H II shows larger bin-to-bin variations in $\gamma^{\rm phys}$ (−0.13, −0.78, +0.40), whereas SIGNALS-H II remains more stable (−0.59, −0.57, −0.75). This likely reflects differences in the luminosity range probed by each survey, with PHANGS extending toward lower-luminosity H II regions, while SIGNALS predominantly samples brighter

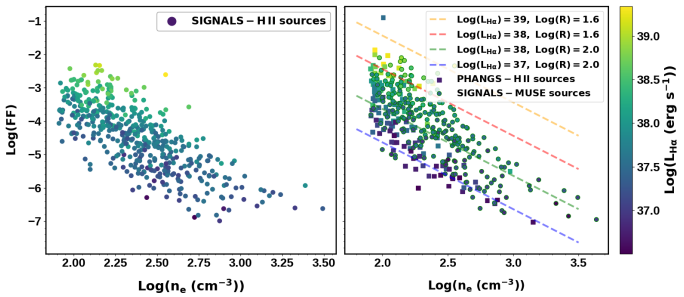


Fig. 10. Log(FF) against the local electron density, color-coded according to the H II region's luminosity. In the left panel, we show the SIGNALS-MUSE regions, and in the right panel the subsample of SIGNALS-MUSE H II regions along with PHANGS-H II regions. The dashed lines show the theoretical log(FF)–log(n$_e$) for selected log(L$_{H\alpha}$) and log(R) values assuming the Strömgren sphere model.

regions. These trends are consistent with the behaviour discussed in Sect. 6.2.1 and may indicate a gradual change in the dominant ionizing sources, from single massive stars to those powered by stellar clusters or associations, across the luminosity range sampled in each catalog, rather than a single universal scaling.

For Groves et al. (2023), comparatively narrow range of local electron densities effectively restricts the sample to a single n_e regime, resulting in a $\gamma^{\rm phys} = -0.09$, representative of the luminous regime probed by this catalog. This behavior likely reflects the specific characteristics of the dataset, including the lack of an explicit DIG subtraction and the resulting impact on the flux distribution.

6.2.2. Relation with the local electron density

We observe a statistically significant negative trend between FF and n_e in SIGNALS-H II and PHANGS-H II catalogs, reflecting the combined influence of density, luminosity, and volume variations on the scatter. Deviations from the exact theoretical exponents are consistent with the impact of density inhomogeneities and the coupling between Hα and the volume.

The fitted slope of the log(FF)–log(n$_e$) relation differs between the two surveys. The SIGNALS-H II sample yields a slope of −2.65 ±0.08. In contrast, the PHANGS-H II sample exhibits a steeper slope of −3.61 ± 0.23. However, when the PHANGS-H II regions are restricted to higher luminosities (log(L$_{H\alpha}$) > 37.5 erg s^{-1}), the slope decreases to −2.7, approaching the value measured for the SIGNALS-H II sample. The steeper slope obtained therefore still reflects the influence of faint, compact regions at the low-luminosity end. The moderate R^2_{fit} values of ∼0.60, indicate that the n_e is the parameter most closely associated with the variance in FF. Within the high-luminosity regime, the classical FF formulation provides a consistent description of the data.

Fig. 10 shows log(FF) versus log(n$_e$) for the two surveys. The figure illustrates the interplay between Hα luminosity, local electron density, and region size, and overplots theoretical relations from the Strömgren sphere model for different combinations of the independent variables.

Fig. 11 shows the log(FF) distribution as a function of the electron density for all the catalogs analyzed in NGC 628. All catalogs display the previously described anti-correlation between log(FF) and log(n$_e$). The SIGNALS-H II catalog spans the widest range of n_e and FF values. The PHANGS-H II and Congiu et al. (2023) samples follow a similar trend but over a

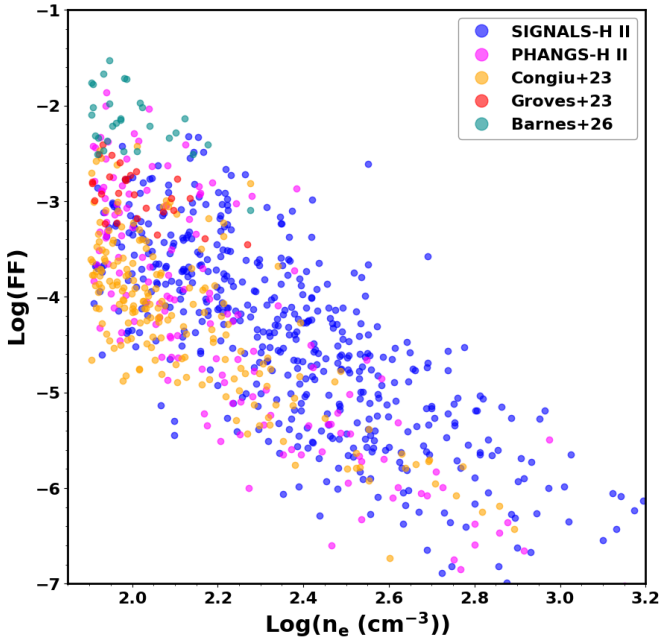


Fig. 11. Distributions of log(FF) vs. log(n_e) in NGC 628 for five catalogs: SIGNALS-H II, PHANGS-H II, and the catalogs of Congiu et al. (2023); Groves et al. (2023) and Barnes et al. (2026).

slightly narrower FF parameter range. The catalog from Groves et al. (2023) mainly populates the high-FF, low-density regime due to its limited range of local electron densities, while Barnes et al. (2026), with local electron densities derived by Groves et al. (2023), occupies a similar low-density regime but at systematically higher FF values. Overall, the log(FF)–log(n_e) relation remains consistent among the different catalogs.

6.2.3. Relation with the luminosity

Hα luminosity presents a strong and statistically significant positive correlation with the FF (left panel of Fig. 12). Here, we use the FF-Hα luminosity relation as a diagnostic tool to assess the regime in which the classical FF formulation provides a physically meaningful description of H II regions. The linear fits yield moderate-to-high R^2_{fit} values (Table 3), yet still leaves ∼40% where the combination of physical conditions contributes to the remaining scatter in the data. The color-coded representation for SIGNALS-H II and PHANGS-H II samples illustrates that similar luminosities span a wide range of local electron densities, highlighting the coupled role of n_e, size, and Hα luminosity in shaping the observed FF values. In the high-luminosity regime, for a given electron density, luminous giant and supergiant H II regions, which are ionized by massive clusters and generally more massive and limited in volume, tend to have higher FF. At lower luminosities, approaching the regime typical of single massive stars, both surveys show increased scatter and systematic departures from the dominant trend. In SIGNALS-H II, enhanced scatter below log($L_{H\alpha}$) < 37.5 erg s^{-1} may be influenced by aspects of the data reduction adopted in this survey. In PHANGS-H II, the regions with log($L_{H\alpha}$) < 36.5 erg s^{-1} reflect the larger diversity of physical conditions, as discussed in Sect. 6.2.1 and Sect. 6.2.2, this defines a coherent low-luminosity regime that is analyzed in Sect. 7.

The right panel of Fig. 12 extends this analysis to all five catalogs. Despite differences in the boundary definition and

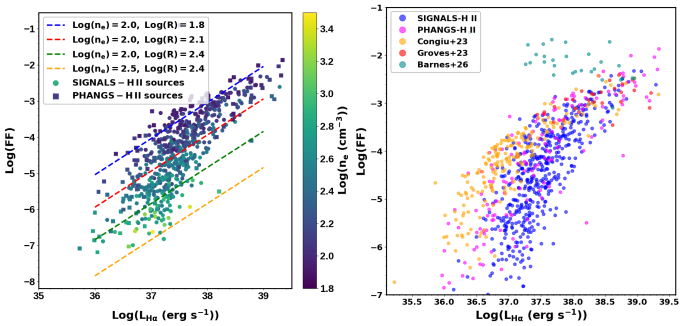


Fig. 12. Log(FF) against the H II region Hα luminosity, log($L_{H\alpha}$), color-coded by the local electron density (left panel). We differentiate the regions by survey: SIGNALS-H II (dots), PHANGS-H II (squares). The dashed lines show the theoretical log(FF)–log($L_{H\alpha}$) relation for selected Log n_e and log(R) values, assuming the Strömgren sphere model. The distributions of log(FF) vs. log($L_{H\alpha}$) in NGC 628 for five catalogs: SIGNALS-H II, PHANGS-H II, and the catalogs of Congiu et al. (2023); Groves et al. (2023) and Barnes et al. (2026) are shown in the right panel.

spatial resolution, the positive FF-$L_{H\alpha}$ trend is consistent in catalogs within the scatter, with Barnes et al. (2026) occupying the high-$L_{H\alpha}$, high-FF end as expected from its high HST-resolution, which resolves compact, high-surface-brightness regions. The systematic offset in median FF between catalogs (Fig. 6) is substantially reduced when comparing regions at fixed luminosity above log($L_{H\alpha}$) > 37.5 erg s^{-1}, confirming that the FF-L relation is more robust against methodological differences than the FF distributions considered alone. After subtracting the methodological contribution, $\eta^{\rm pred} = (\kappa_{05} - 3)/\kappa_{05}$ (Sect. C, the positive correlation between FF and $L_{H\alpha}$ is largely physical in all catalogs with available $\eta^{\rm obs}$ (Table D.1). The physical residuals remain large in SIGNALS-H II and PHANGS-H II ($\eta^{\rm phys}$ = +4.21 and +2.03), indicating that more luminous H II regions intrinsically have higher FF. $\eta^{\rm phys}$ is lower in catalogs with restricted or electron density ranges (Groves et al. 2023; Congiu et al. 2023).

6.2.4. The FF-L/R³ relation: A diagnostic of internal gas structure

Fig. 13 shows log(FF) as a function of the volumetric luminosity density log($L_{H\alpha}/R^3$) for all five catalogs. Despite differences in radius definitions and spatial resolution, spanning nearly two orders of magnitude in characteristic size between Barnes et al. (2026) and the MUSE- and SIGNALS-based catalogs, all datasets follow a common linear relation with similar slope. This behavior follows directly from the definition of the FF, which yields a unit slope in the log(FF)–log($L_{H\alpha}/R^3$) plane under the assumption of fixed n_e, that is, neglecting additional coupling terms such as B (see Appendix D). In this plot, the departure from the definitional slope is a physical result.

This behavior demonstrates that $L_{H\alpha}/R^3$, the volumetric luminosity density of the ionized gas, minimizes the methodological variance introduced by different radius definition criteria, and thus the FF is physically driven by the internal gas structure rather than by the segmentation and definition of boundary method. This makes the volumetric luminosity density a useful quantity to explore how the internal density structure of H II regions varies in different galactic environments.

The primary source of scatter in this representation is the intrinsic variation in electron density in the H II region population. As shown in Eq. (D.4), the dispersion in log(FF) at fixed

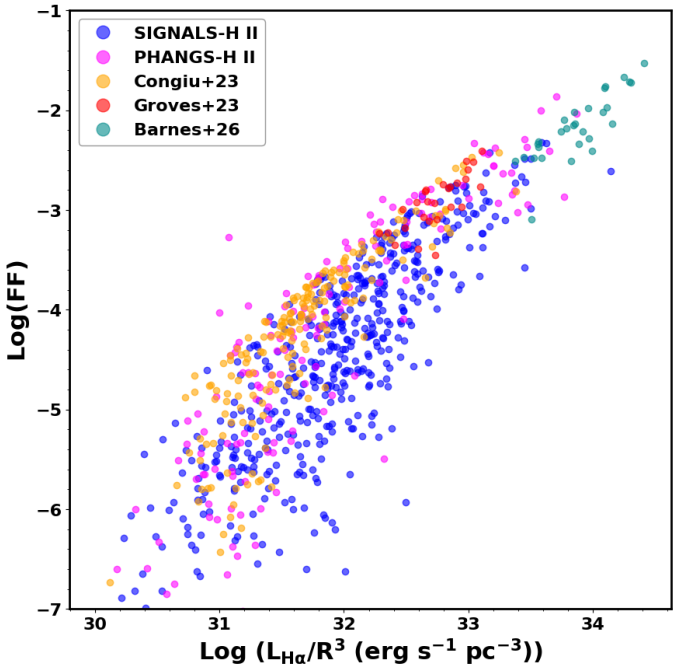


Fig. 13. Log(FF) as a function of the volumetric Hα luminosity density, log($L_{H\alpha}/R^3$), for all five catalogs in NGC 628. The scatter around the relation reflects the variation in n_e^2 in the population of H II regions.

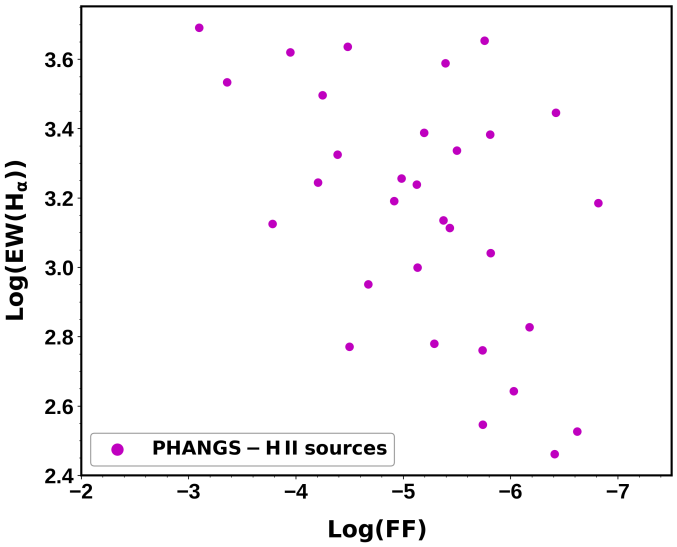


Fig. 14. Log(FF) of H II regions vs. EW(Hα) equivalent width. The age estimated for the PHANGS-H II star forming regions ranges from <3 to 6.3 Myr compared with the Starbust99 model (Leitherer et al. 1999), as described in Ramya et al. (2007) assuming instantaneous-burst with a Salpeter IMF in the mass range 1–10 $M_\odot$. The SIGNALS catalog does not include the EW(Hα) parameter.

$L_{H\alpha}/R^3$ directly traces variations in n_e. It is a direct measure of local electron density heterogeneity. Secondary contributions may include variations in metallicity, electron temperature or nonuniform distribution of clumps, which affect the Hα emissivity coefficient, and departures from the assumption of a uniform clump distribution underlying the classical FF formulation.

Having established the internal consistency and the limitations of the classical FF formulation, we next explore how the FF relates to additional physical tracers. In the following sections, we examine the connection between FF and the *H*α equivalent width, and the dust properties of H II regions, to assess whether FF encodes information about evolutionary stage and interaction with the surrounding ISM.

6.3. Relation with EW(Hα)

Determining the age of H II regions plays a key role in understanding their evolutionary stage. Hα emission traces recent star formation, such that high Hα flux relative to the stellar continuum indicates a recent episode. The equivalent width of Hα, or EW(Hα), represents the ratio of Hα luminosity to the underlying stellar continuum of the young stellar cluster, therefore providing a widely used observational proxy for the age of the ionizing stellar population.

For the PHANGS-H II sample, EW(Hα) values were measured and corrected for the contribution of the underlying galactic stellar population associated with each region. Regions for which the measured stellar continuum was consistent with zero within the uncertainties were excluded from the analysis. This requirement reduces the available sample to 18 H II regions. This result should therefore be regarded as exploratory, given the small number of regions available. Future studies will extend this analysis to other nearby galaxies.

Fig. 14 presents the distribution of log(FF) as a function of log(EW(Hα)) for the PHANGS-H II regions. When compared with Starburst99 instantaneous-burst models (Leitherer et al. 1999), the inferred ages for the sample span from <3 to 6.3 Myr. We find a positive correlation between the log(FF)–log(EW(Hα)) relation in PHANGS H II regions (ρ = 0.56, p-value <0.001), suggesting that high-FF regions tend to show high EW(Hα), characteristic of younger populations.

However, we note that the two quantities are not fully independent, as they share an implicit dependence on the Hα luminosity, and therefore part of the observed correlation may be driven by this underlying covariance. Nonetheless, other independent tracers of the H II region evolutionary state have been proposed in the literature (e.g., excitation, kinematics) that could be used in future work.

We identified three broad regimes in this parameter space: (i) high EW(Hα) and high FF (log(EW(Hα)) $\gtrsim$ 3; Log(FF) $\gtrsim -3$), corresponding to very young, compact H II regions with high filling factors; (ii) intermediate log(EW(Hα)) and moderate FF (log(EW(Hα)) $\lesssim$ 3 to 2; Log(FF) ~ -3 to -5), associated with more evolved regions where the ionizing flux remains significant but the nebular gas becomes more spatially extended; and (iii) low log(EW(Hα)) and low FF (log(EW(Hα)) $\lesssim$ 1.5; Log(FF) $\lesssim -5$), characteristic of older or more evolved H II regions, in which nebular emission decreases relative to the stellar continuum and the ionized gas occupies a smaller fraction of the volume. We note that this regime is distinct from the low-luminosity, compact regions discussed in Sect. 6.2.1, which exhibit low FF values due to the nature of their ionizing sources. Extending this analysis into the single massive star regime would require larger samples of faint, compact H II regions observed at higher spatial resolution.

However, the evolutionary sequence of giant and supergiant H II regions does not necessarily follow a simple linear sequence, as their ionization could be driven by stellar associations or clusters comprising multiple stars at different evolutionary stages, which can introduce additional scatter in the FF–EW(Hα)

relation. Further work is underway to study this sequence for a larger sample of galaxies.

6.4. Relation with the PAH-to-dust ratio

Using the enhanced angular resolution provided by JWST/MIRI, reaching scales of a few tens of parsecs, we are able to investigate the presence of small-scale correlations between the FF and the local dusty environment of H II regions.

We performed circular aperture photometry for H II regions of SIGNALS and elliptical aperture for H II regions of PHANGS on each MIRI image of NGC 628 (F770W, F1130W, and F2100W), using the Photutils Python package, adopting the calibration coefficients listed in Table 4 of Lee et al. (2023) and the conversion factors PHOTMJSR and PHOTUJA2 provided in the FITS files. Photometric measurements were carried out for all locations corresponding to the sample of PHANGS-H II and SIGNALS-H II regions within the MIRI FoV. For each region, the flux was computed by summing the contribution of all pixels (in $\mu Jy/arcsec^2$) enclosed within the projected area of each H II region.

We did not regrid the infrared images to the astrometric grid of the PHANGS-MUSE NGC 628 data in order to avoid introducing potential additional errors from interpolation effects. Instead, we rescaled and reprojected the apertures to match the geometrical properties (size, orientation, and position) of each H II region as defined in the corresponding catalogs. As a consequence, we adopted a variable aperture radius, matched to the equivalent radius of each H II region, rescaling and reprojecting it individually to match the resolution and orientation of each image. In all cases, the adopted aperture size exceeds the instrumental PSF.

We focused on a subset of 158 H II regions from the SIGNALS-H II sample (approximately 33% of the sample) that are covered by the JWST footprint and exhibit nonzero PAH emission. This subsample is hereafter referred to as SIGNALS-JWST. For the PHANGS-H II catalog, 105 regions (about 71% of the 147 selected regions) fall within the JWST coverage and are hereafter labeled PHANGS-JWST. The distribution of radii, $H\alpha$ luminosities, and FF of the regions within the JWST mask closely mirrors those obtained for the MUSE samples, as discussed in Section 6.2 and we illustrate in Fig. G.6.

Following Chastenet et al. (2023), we quantified the relative contribution of PAHs using the combined emission in the 7.7 μm and 11.3 μm bands, normalized by the thermal dust emission traced by the 21 μm band:

$$R_{\rm PAH} = \frac{F_{F770W} + F_{F1130W}}{F_{F2100W}} \tag{15}$$

We analyzed each catalog independently to investigate the correlation between R_{PAH} and $\log(L_{H\alpha}/R^3)$. This will minimize potential biases from differences in the H II region detection techniques and/or emission line flux measurement methods.

In Fig. 15, the PHANGS-JWST and SIGNALS-JWST samples suggest the presence of two distinct regimes in the relation between the $\log(R_{PAH})$–$\log(L_{H\alpha}/R^3)$, with a noticeable change in the slope. To characterize the global behavior of the relation between $\log(R_{PAH})$ and the $\log(L_{H\alpha}/R^3)$, we applied a locally weighted scatterplot smoothing (LOWESS) regression to the data ($f = 0.25$). LOWESS is a nonparametric method that estimates the underlying trend by performing local weighted linear fits over neighboring subsets of the data, without assuming a specific functional form for the relation.

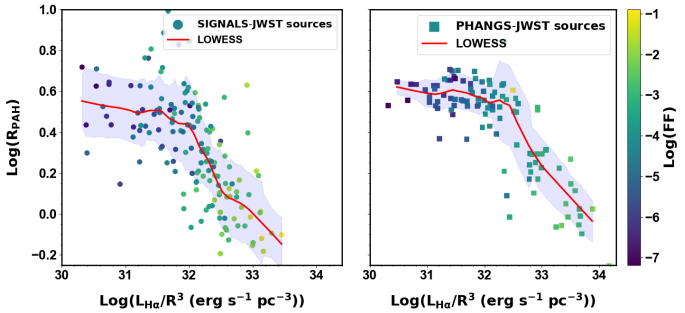


Fig. 15. Correlation of the PAH abundance traced with R_{PAH} with $\log(L_{H\alpha}/R^3)$ in SIGNALS-JWST (left panel) and PHANGS-JWST H II regions (right panel). The red curve shows a LOWESS smoothing of the data, highlighting the global trend.

The resulting LOWESS-smoothed relation reveals a clear change in slope around $\log(L_{H\alpha}/R^3)$ ~32 erg s^{-1} pc^{-3} in both samples. To quantify the range of this transition, we performed bootstrap resampling of the datasets with 1000 iterations. We obtained distributions of transition values with mean and standard deviation of $\log(L_{H\alpha}/R^3)$ =31.64 ± 0.16 for the SIGNALS–JWST sample and $\log(L_{H\alpha}/R^3)$ = 32.03 ± 0.20 for the PHANGS–JWST sample. The global shape of the two samples appears to be consistent within the uncertainties. The comparatively large dispersions indicate that the transition is not a sharp breakpoint but rather a gradual change over an extended range in FF. Given the small offset and comparable dispersions, we defined a common transition regime as $31.5 \leq \log(L_{H\alpha}/R^3) \leq 32.2$, encompassing the bulk of both distributions. This choice enables a consistent comparison while acknowledging that the transition does not occur at a single universal value. The H II regions within the transition emissivity range span hydrogen-ionizing photon rates of $Q_H \sim 10^{49} - 10^{50}$ s^{-1}, with median values of $Q_H \sim 2.58 \times 10^{49}$ s^{-1} (SIGNALS-JWST) and $Q_H \sim 1.56 \times 10^{49}$ s^{-1} (PHANGS-JWST), calculated using Case B relation of Osterbrock & Ferland (2006) ($Q_H = 7.35 \times 10^{11} L H\alpha$). These values are consistent with the threshold reported by Egorov et al. (2023b), above which $Q_H \sim 10^{50}$ s^{-1} PAH destruction becomes efficient. The transition identified in the $\log(R_{PAH})$-$\log(L_{H\alpha}/R^3)$ relation thus corresponds to the regime where the ionizing output of H II regions approaches this critical rate, marking the onset of efficient PAH photodissociation rather than its fully established phase. This interpretation is further supported by the upper envelopes of both samples, which reach $Q_H \sim 10^{50}$ s^{-1} at the high-volumetric Hα luminosity density end of the transition range. Future studies applying this analysis to additional nearby galaxies will provide a more precise determination of the characteristic volumetric Hα luminosity density at which PAH depletion begins.

The FF values of H II regions within the transition emissivity range are consistent between the two catalogs. For SIGNALS-JWST, the defined transition interval contains 51 regions with a median FF of log(FF) = –4.58 and an interquartile range of [–5.01, –4.09]. For PHANGS-JWST, the transition interval contains 27 regions with a median of log(FF) = −4.28 [−4.80, −4.02], with a median offset of 0.22 dex, well within the standard deviation of ~ 0.59 and 0.63 dex respectively present in both samples. Despite the difference in the transitional range in the two catalogs, the FF distributions are statistically consistent, indicating that the physical characterization of the transition is robust.

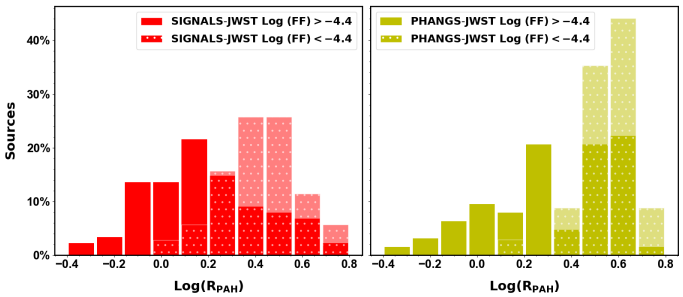


Fig. 16. Histogram showing the number of regions with the PAH abundance traced with R_{PAH} as a function of the FF $\lesssim -4.4$ in SIGNALS-JWST (left panel) and PHANGS-JWST (right panel) H II regions contained in the FoV of the JWST. The vertical axis represents the total number of regions in percentage.

For simplicity, we adopt a first-order approximation of the transition point at log(FF) ~ -4.4. We used Spearman's correlation coefficient to test whether R_{PAH} and log(FF) are monotonically associated, regardless of the functional form of the relation. For PHANGS-JWST, 42 regions with log(FF) < –4.4 show no significant correlation (ρ = 0.12, p-value = 0.77), with a characteristic value of log(R_{PAH}) = 0.60 ± 0.05. In contrast, 63 regions with log(FF) > –4.4 exhibit a strong statistically significant negative correlation (ρ = –0.69, p-value = < 0.001), indicating that the relation strengthens above this threshold. A similar behavior is found for the SIGNALS-JWST sample. 70 regions with log(FF) < –4.4 do not display a significant correlation (ρ = –0.18, p-value = 0.13) with log(R_{PAH}) = 0.47 ± 0.23, whereas 88 regions with log(FF) > −4.4 show a significant negative correlation (ρ = –0.54, p-value = < 0.001).

Overall, these results indicate that in both samples, the strength of the log(R_{PAH})-log(FF) correlation increases for regions above log(FF) value of –4.4, revealing a clear transitional change in behavior around this threshold. The subset of high-luminosity H II regions with larger FF tends to exhibit lower PAHs ratios, whereas fainter regions show higher PAH abundances. This behavior could be understood as a consequence of the weaker ionizing radiation field, which is insufficient to efficiently produce photodissociation and destruction of PAHs. However, we should bear in mind that part of this trend may also be driven by observational effects in marginally resolved regions.

This result is fully consistent with the values reported by Egorov et al. (2023b). H II regions with $f_{ion} < 0.95$ are associated with lower log(FF) and show no evidence of PAH destruction. For the PHANGS-JWST and SIGNALS-JWST samples R_{PAH} distributions of Fig. 16, we test the equality of variances using Levene's test rejected the null hypothesis of equal variances for PHANGS-JWST (p-value = 0.001), but not for SIGNALS-JWST (p-value = 0.71). However, a Shapiro–Wilk test indicated at least one subsamples deviates significantly from normality. Therefore, we used the nonparametric Mann–Whitney U test to assess differences between the R_{PAH} distributions in the two FF regimes in the two subsamples.

In the PHANGS–JWST sample, a Mann–Whitney U test reveals a statistically significant difference between the two FF regimes (U = 510, p-value < 0.001). The effect size is large (rank-biserial correlation $r_{rb} = -0.62$), with a "Common Language Effect Size" (CLES) ~0.81, indicating that values in the low-FF sample exceed those in the high-FF sample in approximately 76% of pairwise comparisons. A consistent result is obtained for the SIGNALS–JWST sample, where the Mann–Whitney U test also indicates a highly significant difference between the two populations (U = 1052, p-value <0.001). The effect size remains large ($r_{rb} \approx -0.66$), with an effect size of CLES of ~0.83, confirming that regions in the low-FF regime systematically exhibit higher PAH ratios and that the effect does not depend on the specific catalog.

Finally, a qualitative visualization of this result is suggested in Fig. 17 by the likely association between regions with the lowest FF and areas of enhanced PAH ionized emission. We combined PHANGS-Hα with PHANGS-JWST/MIRI F770W image to produce a false-color composite image at their native resolutions. We overlay the log(FF) values of individual H II region to highlight those with log(FF) < −4.4 which are expected to most clearly exhibit the observed trends.

Lower log(FF) values (white circles for SIGNALS-JWST and orange ellipses for PHANGS-JWST) appear preferentially located within or near the red regions in the MIRI F770W image, which exhibit systematically higher R_{PAH} ratios (Fig. 15). This is consistent with less efficient PAH processing in regions characterized by more porous ionized-gas structures.

Assuming that regions with very high FF are, on average, powered by younger stellar populations, the intense radiation fields of central massive stars may initially produce small PAHs and subsequently destroy them. At later evolutionary stages, stellar winds and mechanical feedback can disperse the ionized gas, while reducing the efficiency of PAH destruction. This evolutionary scenario is not expected to be uniform, as it ultimately depends on the particular physical conditions of the ISM in each region because the mechanisms governing PAH destruction, survival, and migration remain poorly constrained (e.g., García-Bernete et al. 2022; Hrodmarsson et al. 2023).

7. Discussion

The main goal of this study is to establish a foundation for future investigations of H II regions at similar spatial resolution, exploring how FF reflects the internal structure and connects to the physical and chemical properties of extragalactic star-forming regions in NGC 628. We consider the FF as a global parameter, and acknowledge that it encompasses several distinct physical components (Sect. 1), while spherical ionization-bounded approximations together with uniform density and spherical symmetry, represent a first-order simplification.

The typical spatial resolutions of the SIGNALS and the PHANGS-MUSE surveys in NGC 628 do not allow us to resolve substructures such as rings, shells, or cavities within the analyzed H II regions. We selected a total of 622 H II regions from the SIGNALS and the PHANGS-MUSE H II regions survey. After applying a consistent homogenization of the physical parameters and density measurements in the two surveys, we found that the derived FF values showed very good agreement. The statistical consistency between the distributions confirms the equivalence of the SIGNALS and PHANGS-MUSE datasets for this analysis and provides a robust basis for our subsequent results. We also compared our results with published H II region catalogs of the same galaxy based on PHANGS-MUSE and PHANGS-MUSE HST data. These catalogs use different segmentation pipelines and boundary definitions. We demonstrated that the apparent slope of the observed Hα luminosity–R relation is not universal, but rather a methodological effect reflecting the position of the radius definition threshold relative to the intrinsic profile scale of each region (Sect. 4). The cross-catalog analysis demonstrated in Fig. 1 confirms that this scatter is not an effect

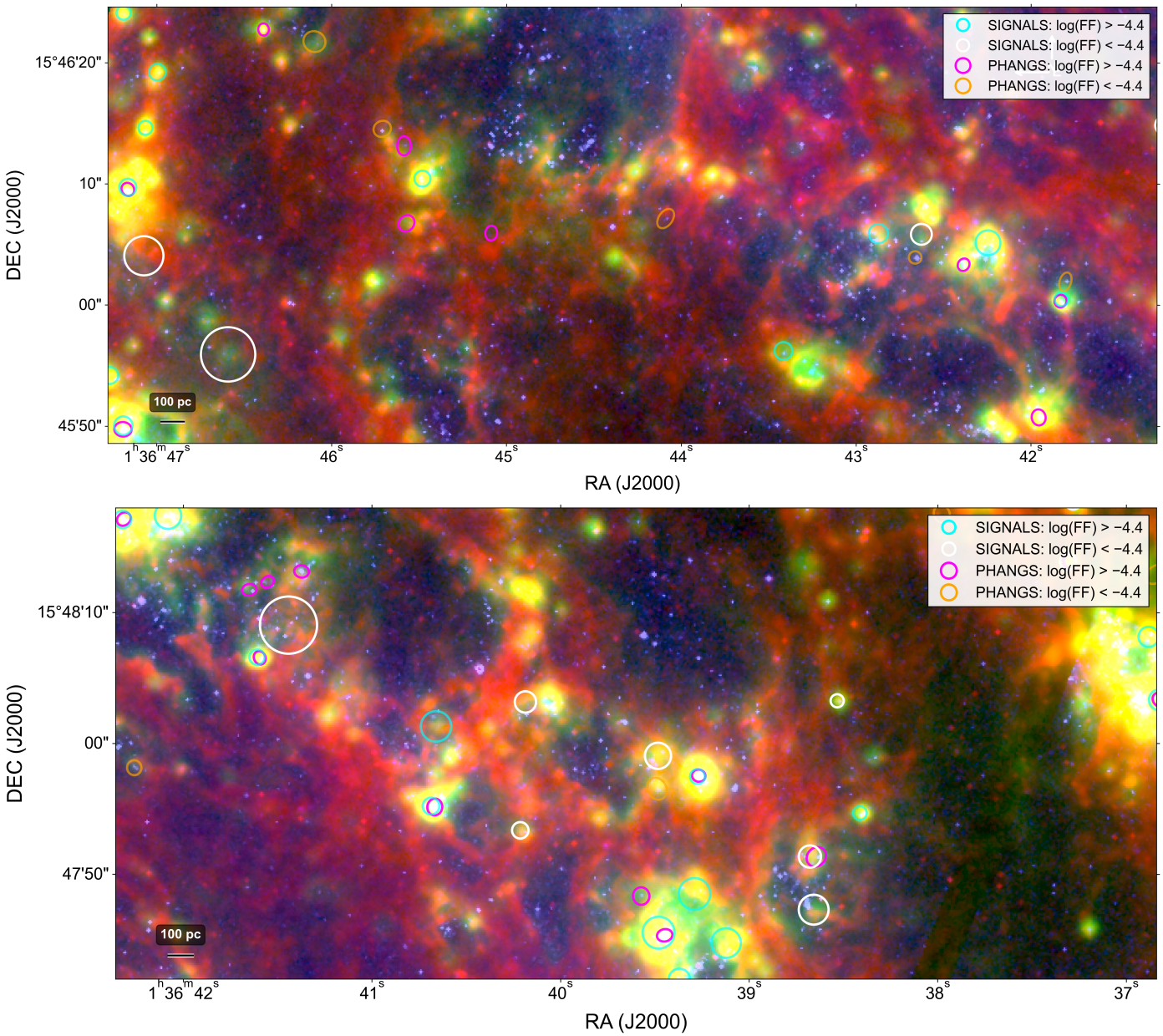


Fig. 17. Close-up of the central southwest area of the galaxy NGC 628 showing log(FF) of each H II region in a local RGB composite image. The circles measured are plotted in light blue for SIGNALS-JWST regions with log(FF) > −4.4 and as white circles for regions with log(FF) < −4.4. In PHANGS-JWST, magenta ellipses illustrate regions with log(FF) > −4.4 and orange ellipses those with log(FF) < −4.4. Red corresponds to the MIRI F770W image, green to the PHANGS-MUSE Hα image and blue to the HST F435W image of NGC 628. Low-log(FF) regions are preferentially associated with red MIRI F770W emission, whereas high-log(FF) regions appear associated within the brightest PHANGS-MUSE Hα structures.

of the boundary definition but reflects physical variation in n_e^2 in the population.

Luminous giant and supergiant H II regions show the expected decrease in FF with increasing volume (Gutiérrez & Beckman 2010; Cedrés et al. 2013), consistent with the definitional scaling. This behavior indicates that larger regions are characterized by more widely spaced clumps and, consequently, by a larger fraction of low-density gas occupying the available volume (Hunt & Hirashita 2009). However, the variance explained by volume alone is low ($R^2_{fit} \sim 0.16$–0.21), indicating that volume alone cannot explain the intrinsic scatter of FF within this group of regions.

The local electron density exhibits a strong anti-correlation with FF, consistent with the n_e^{-2} slope in the FF definition. SIGNALS-H II regions show excellent agreement with the expected slope, while PHANGS-H II regions display more scatter, particularly in low-luminosity and compact regions. Most of the brightest giant H II regions are found in the low-density regime, with lower FF (longer optical paths) than fainter giant H II regions. This suggests that in fainter regions, the ionized gas is distributed in a more porous and inhomogeneous medium with greater spacing between condensations. The increased separation between inhomogeneities could result in a substantial low-density volume, allowing ionizing photons to escape more efficiently from central stars through leaky paths, as suggested

by Belfiore et al. (2022); Teh et al. (2023). This effect supports a scenario in which the internal geometry and possibly the photon escape fraction play key roles in regulating the ionized gas structure (Giammanco et al. 2004). This process could, in turn, contribute to enhanced DIG emission in the surrounding environment.

Hα luminosity correlates positively with FF, consistent with the analytical relation $FF \propto L/(n_e^2 R^3)$. Luminous H II regions, typically ionized by stellar associations or clusters containing multiple OB stars (Santoro et al. 2022; Scheuermann et al. 2023), exhibit higher FF values, consistent with ionized gas confined to dense clumps embedded within a low-density medium. At lower luminosities, however, systematic departures from the dominant trends emerge. Regions with $\log(L_{\mathrm{H}\alpha}) \lesssim 37$–$37.5$ erg s^{-1} show increased scatter in the FF–$L_{\mathrm{H}\alpha}$ relation, fall below the FF–R relation reported by Cedrés et al. (2013), and drive the steeper FF–n_e slope measured for PHANGS-H II (-3.61 ± 0.23); excluding them recovers a slope of ~ -2.7, consistent with the SIGNALS-H II sample. In the FF–$L_{H\alpha}$ relation, increased scatter and departures below the dominant trend appear at the faint end in both surveys. These objects comprise compact regions ($R < 50$ pc) together with a population of somewhat larger nebulae, and their behavior appears largely independent of local ISM conditions owing to the DIG-corrected measurements. Their inferred ionizing luminosities are consistent with excitation by single late O-type or early B-type stars, or by low-mass clusters (Lee et al. 2011), making them analogous to classical Galactic H II regions despite being located in external galaxies (Kennicutt 1984). The transition does not correspond to a sharp luminosity threshold, but rather to the luminosity range over which deviations from the classical FF scaling become apparent, in agreement with the regime change identified by Lee et al. (2011) around $\log(L_{\mathrm{H}\alpha}) \lesssim 37.1$ erg s^{-1}. Overall, the classical FF framework remains valid throughout the full luminosity range we probed. In the luminous regime, regions with higher FF values, indicative of reduced interclump spacing and a larger fraction of ionized gas filling the nebular volume, tend to exhibit lower local electron densities than more extended, lower-FF regions. We adopted $\log(\mathrm{L_{H\alpha}}) \lesssim 37$–$37.5$ erg s^{-1} as an approximate luminosity range over which the transition between regimes becomes apparent, in line with the threshold reported by (Lee et al. 2011) around $\log(\mathrm{L_{H\alpha}}) \lesssim 37.1$ erg s^{-1}. This range does not represent a sharp threshold, but rather reflects that the departure becomes apparent at different luminosities depending on the scaling relation considered.

Strong radiation pressure in luminous H II regions drives gas expansion, concentrating material into dense shells where the [S II] lines probe the local electron density. In this regime, the rms-based FF provides a meaningful description of the nebular geometry and closely approximates the idealized Strömgren-sphere picture. At lower luminosities, however, ionized structures become increasingly affected by sampling and resolution effects, with filaments and condensations contributing significantly to the density field. Consequently, FF no longer acts as a purely volumetric measure but instead reflects the underlying density distribution. A more physical description would therefore require approaches beyond the classical Osterbrock framework and purely statistical PDF-based measures (Spangler et al. 2019), such as fractal analyses or MHD simulations capable of deriving FF directly from the gas dynamics. A detailed exploration of these alternatives is beyond the scope of this work.

One of the most significant results of this study is the linear alignment of all five H II regions catalogs in the $\log(\mathrm{FF})$–$\log(\mathrm{L_{H\alpha}}/\mathrm{R}^3)$ plane (Fig. 13). Despite nearly two orders of magnitude difference in radii between Barnes et al. (2026) and the SIGNALS- and MUSE-based catalogs, and despite the varying criteria used to define H II region boundaries, all datasets fall on the same relation. This is nontrivial: the scaling relation between $\log(\mathrm{FF})$ and $\log(\mathrm{L_{H\alpha}}/\mathrm{R}^3)$ does not imply that catalogs defining R via absolute flux thresholds, profile truncation, or HST-based fragmentation would coincide. The usefulness of the $\log(\mathrm{FF})$–$\log(\mathrm{L_{H\alpha}}/\mathrm{R}^3)$ plane lies in isolating the role of the electron density term. The residual scatter around this relation reflects true variations in n_e^2 rather than differences in segmentation. Future work will extend this analysis beyond NGC 628 to a larger sample of galaxies (Aragüete Riesco et al., in prep.). This demonstrates that $\mathrm{L_{H\alpha}}/\mathrm{R}^3$, the volumetric H$\alpha$ luminosity density absorbs methodological differences and measures the effective emission measure per unit volume of the emitting clumps. Furthermore, correlations involving $\mathrm{L_{H\alpha}}/\mathrm{R}^3$ are comparable in all catalogs, provided that the second observable does not contain an explicit dependence on the adopted region size.

As the central cluster's ionizing output declines, the EW(Hα) progressively decreases (e.g. Kewley et al. 2015; Mármol-Queraltó et al. 2016; Mingozzi et al. 2020). In this context, the observed decrease in EW(Hα) toward lower FF values may indicate an advanced evolutionary stage. These preliminary results provide a coherent picture linking internal gas structure and evolutionary stage: young compact regions with high EW(Hα) and high FF; intermediate regions with extended nebulae and moderate EW(Hα); and older regions with low EW(Hα) and low FF, with inferred ages spanning from less than 3 to 6.3 Myr. However, given the shared dependence on Hα luminosity, the observed FF–EW(Hα) trend cannot be unambiguously interpreted in terms of evolutionary sequence alone.

However, we caution against a definitive evolutionary interpretation. The classical Strömgren framework assumes uniform density and spherical symmetry, an unrealistic simplification in real H II regions (Castaneda et al. 1992; Copetti et al. 2000), whereas real H II regions are often powered by multiple OB associations at different evolutionary stages. This would introduce additional scatter in any FF–EW(Hα) relation and further limits its interpretation as a strict evolutionary sequence. In addition, our analysis is exploratory, based on a limited number of H II regions with reliable continuum measurements, and relies on EW(Hα) as an indirect age proxy.

Elevated EW(Hα) values in bright H II regions may also reflect metallicity effects (Leitherer et al. 1999) or leakage of ionizing photons (Zurita et al. 2002). A definitive evolutionary interpretation would require independent age constraints beyond the EW(Hα) proxy, for instance from stellar cluster catalogs. Furthermore, the assumption of a uniform density of ionized gas clumps is an additional simplification that may not hold in turbulent or highly inhomogeneous regions (Copetti et al. 2000).

We examined the relation between FF and PAH emission. The transition in PAH abundance identified in the $\log(\mathrm{R_{PAH}})$-$\log(\mathrm{L_{H\alpha}}/\mathrm{R})$ relation corresponds to H II regions with volumetric Hα luminosity density, $\log(\mathrm{L_{H\alpha}}/\mathrm{R}^3) \approx 31.76$–$32.11$ erg s^{-1} pc^{-3}, ionizing photon rates $\log(\mathrm{Q_H}) \approx 49.03$–$49.48$ s^{-1}), and $\log(\mathrm{FF}) \approx -5.01$ to -4.02. The consistency of these three independent diagnostics in two catalogs, built from different instruments and radius definition methods, confirms that this physical regime marks the onset of efficient PAH photodissociation in H II regions of NGC 628, in agreement with the critical threshold $\log(\mathrm{Q_H}) \gtrsim 10^{50}$ s^{-1} reported by Egorov et al. (2023b).

Within the luminous giant and supergiant H II region regime, characterized by high FF values, we find a clear dependence between the PAH-to-dust fraction and FF. PAHs are highly

susceptible to Lyman-continuum radiation, undergoing photoionization and photodissociation that progressively reduce their PAH fraction (Dale et al. 2025). High-FF regions, which are typically the most luminous, provide the intense radiation fields required for efficient PAH destruction. In addition, stellar winds and radiation pressure from massive O/B stars can evacuate the central regions and compress dust and gas into shell-like structures where PAH emission is preferentially detected (Watson et al. 2008). Our results suggest that lower-luminosity H II regions exhibit higher PAH-to-dust ratios, indicating less efficient destruction of PAHs and carbonaceous grains, consistent with the findings of Egorov et al. (2023b, 2025). Beyond reduced photodestruction, evolutionary effects may also contribute: the shattering of larger carbonaceous grains and the preferential destruction of VSGs can enhance the relative PAH abundance in older H II regions (Jones et al. 1996; Galliano et al. 2008; Micelotta et al. 2010; Seok et al. 2014). Our observations support this scenario in which PAH survival and replenishment become increasingly important as the ionizing radiation field weakens with time.

In the case of very low-luminosity, small H II regions, we observe an almost flat dependence of $\log(R_{PAH})$ on log(FF). This suggests that UV radiation in these regions is insufficient to efficiently destroy PAHs, and that secondary mechanisms unrelated to ionizing radiation may play a more important role in maintaining the PAH population (Reizer et al. 2022). Furthermore, in small H II regions, the measured R_{PAH} values are significantly affected by observational biases. As discussed by Egorov et al. (2025), approximately 45% of these measurements are overestimated by more than 10% due to the difficulty of fully isolating the ionized gas from the surrounding diffuse ISM and PDRs at the current resolution. This contamination, compounded by line-of-sight projection effects, may explain why smaller regions often exhibit R_{PAH} values closer to those of the diffuse interstellar medium. However, in NGC 628, the mean R_{PAH} of the diffuse ISM is $3.72^{+4.50}_{-2.84}$ (Sutter et al. 2024). We consistently find higher values in small, low-luminosity H II regions, suggesting that additional mechanisms, such as accretion-driven processes leading to the formation of small PAHs, may help sustain the PAH population and partially compensate for reduced radiative destruction. To test the impact of this bias, we propagate the expected overestimation to the median R_{PAH} for regions with log(FF) < −4.4. If 45% of these regions are overestimated by a factor of 1.10 (Egorov et al. 2025), the corrected median becomes $R_{PAH,corr} = R_{PAH,obs}/1.045$, implying $\Delta R_{PAH} \approx -0.02$ dex. This shift is smaller than the intrinsic dispersion of the low-FF subsample ($\sigma_{R_{PAH}} = 0.11$ dex in SIGNALS–JWST and $\sigma_{R_{PAH}} = 0.22$ dex in PHANGS–JWST) and below the median offset between FF regimes ($\Delta R_{PAH} \approx 0.18$–$0.25$ dex; Fig. 16), indicating that the observational bias cannot explain the elevated R_{PAH} values in low-FF regions.

Overall, our results show that the FF is a useful diagnostic of the internal structure and evolutionary stage of extragalactic H II regions. The apparent slope of the $L_{H\alpha}$–R relation depends on whether the boundary definition is tied to the absolute flux level or to the local profile structure of each region providing a framework for cross-catalog comparisons. The tight alignment of all five catalogs in the log(FF) –$\log(L_{H\alpha}/R^3)$ plane indicates that the volumetric luminosity density offers a convenient common representation of the $L_{H\alpha}$–R coupling when combined with FF estimates, reducing the impact of methodological differences in radius definitions.

The applicability of the FF framework throughout the full luminosity range probed here provides a basis for exploring variations in H II region structure and dust properties in diverse extragalactic environments (Aragüete Riesco et al., in prep.).

8. Conclusions

We presented a detailed census of extragalactic H II regions in NGC 628, providing homogeneous measurements of luminosity, emission line fluxes, and radii by combining 475 H II regions from the SIGNALS survey and 147 regions from PHANGS-MUSE in NGC 628. We provided measurements of the Hα luminosity, emission line fluxes, electron densities, radii, and FF in the two surveys and extended the comparison to three published H II region catalogs based on different segmentation methods. Our analysis confirmed a close connection between the FF and several fundamental properties of H II regions, highlighting its diagnostic value for probing their internal structure. We also investigated the connection between H II regions and the PAH-to-dust ratio, offering new insights into the interplay among star formation, ionized gas, dust, and the interstellar medium. Our analysis yields several key findings that we list below:

- The apparent slope of the $L_{H\alpha}$-R relation does not appear to be a universal physical constant, but is an observable that can be affected by the coupling between the boundary definition and the flux distribution of each region. Assuming a Gaussian surface brightness H II region profile model, we demonstrated that absolute-threshold methods produce steeper effective slopes than profile-truncated methods. This framework distinguishes physical and methodological effects in Hα luminosity–radius relations;
- We showed that the SIGNALS, PHANGS–MUSE, and PHANGS–MUSE/HST Hα catalogs agree very well in the log(FF) –$\log(L_{H\alpha}/R^3)$ plane. The residual scatter primarily reflects intrinsic variations in n_e^2 in the H II region population.

  The volumetric Hα luminosity density, $L_{H\alpha}/R^3$, reduces the covariance between $L_{H\alpha}$ and R introduced by the adopted region definition, and it provides a consistent way to compare H II region properties in different catalogs in slope space. Physically, it traces the effective emission measure per unit volume of the emitting clumps, $L_{H\alpha}/R^3 \propto n_e^2 \times$ FF. In this sense, $L_{H\alpha}/R^3$ constitutes a useful quantity for cross-catalog comparisons when it is correlated with observables that are independent of the adopted region size;
- The FF scaling relations with volume, electron density, and Hα luminosity are globally consistent with expectations from a Strömgren-like framework in the luminous regime. However, the scatter around each relation carries physical information about the internal gas structure of H II regions and reflects secondary dependences;
- Low-luminosity H II regions, likely powered by single massive stars, appear to depart from the trends followed by giant and supergiant regions in all three FF scaling relations. These differences are primarily driven by observational sampling and resolution effects and not by a breakdown of the classical FF formulation. The transition occurs over the range $\log(L_{H\alpha}) \sim 36.5$–$37.5$ erg s^{-1}, depending on the relation considered;
- The negative physical residual γ^{phys} in the log(FF)–$\log(R^3)$ relation reflects that more extended regions host lower n_e^2, so that the increasing volume-to-luminosity ratio and the dilution of ionized gas act together to reduce FF beyond what is predicted by the definitional and coupling effects alone.

The log(FF)–log($L_{H\alpha}$) relation indicates that more luminous regions genuinely occupy a larger ionized fraction of their volume;
- In an exploratory analysis, the log(FF)–log(EW(Hα)) relation suggests that regions with high log(EW(Hα)) (interpreted as younger ages) tend to show higher log(FF), while intermediate and lower log(EW(Hα)) regions span progressively lower log(FF) values. The inferred ages range from $\lesssim$3 to ~6.3 Myr, although a robust evolutionary interpretation requires larger samples and independent age constraints beyond the EW(Hα) proxy. These trends might be consistent with a decrease in FF with H II region age;
- In NGC 628, the log(R_{PAH})–log($L_{H\alpha}/R^3$) relation shows a transitional change in the slope at log($L_{H\alpha}/R^3$) $\approx$ 31.76–32.11 erg s^{-1} pc^{-3}, below which H II regions preserve higher PAH abundances, indicating less efficient photodestruction in more diffuse, evolved environments. Above this, R_{PAH} decreases steeply with increasing $L_{H\alpha}/R^3$, consistent with efficient PAH destruction by the intense radiation fields of luminous H II regions. The H II regions at the transition are characterized by ionizing photon rates log(Q_H) $\approx$ 49.03–49.48 s^{-1} and log(FF) $\approx$ −5.01 to −4.02.

Taken together, these trends can be indicative of a progressive change in the H II region properties, where a lower FF is associated with more extended and diffuse ionized structures that appear to be more favorable to PAH survival. Further investigations extending this analysis to other galaxies and higher spatial resolutions are currently underway.

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Appendix A: Analytical derivation of the log($L_{H\alpha}$)-log(R) relation

A.1. Absolute-threshold radius definition methods

The Hα surface brightness $\Sigma_{H\alpha}$ is related to the emission measure EM = $\int n_e^2\, dl$ through:

$$\Sigma_{H\alpha} = \frac{\alpha_{H\alpha}^{\rm eff}\, h\nu_{H\alpha}}{4\pi}\,\mathrm{EM} = 1.09\times10^{-6}\,\mathrm{EM} \quad (\mathrm{erg\,s^{-1}\,cm^{-2}\,sr^{-1}}) \tag{A.1}$$

where $\alpha_{H\alpha}^{\rm eff} = 1.17\times10^{-13}$ cm^3 s^{-1} at $T_e = 10^4$ K (Case B; Osterbrock & Ferland 2006) and EM is expressed in pc cm^{-6}. This relation is exact under uniform T_e, which is implicit in adopting a fixed $\mathrm{EM_{min}}$ threshold for all regions (Groves et al. 2023).

We can model each H II region as having a Gaussian surface brightness profile:

$$\Sigma(r) = \Sigma_0\, \exp\left(-\frac{r^2}{2\sigma^2}\right) \tag{A.2}$$

where an absolute flux threshold $\Sigma_{\rm min}$ truncates the profile at the radius where $\Sigma(R_{\rm limit}) = \Sigma_{\rm min}$:

$$R_{\rm limit} = \sigma\sqrt{2\ln\left(\frac{\Sigma_0}{\Sigma_{\rm min}}\right)} \tag{A.3}$$

Since $\Sigma_{\rm min}$ is fixed and Σ_0 varies, $R_{\rm limit}$ grows with the peak surface brightness of each region. The total luminosity integrated up to $R_{\rm limit}$ is:

$$L_{H\alpha} = 2\pi\sigma^2\Sigma_0\left[1-\exp\left(-\frac{R_{\rm limit}^2}{2\sigma^2}\right)\right] \tag{A.4}$$

Substituting the threshold condition, we get:

$$\mathrm{L_{H\alpha}} \approx 2\pi\sigma^2\Sigma_{\rm min}\exp\left(\frac{R_{\rm limit}^2}{2\sigma^2}\right) \tag{A.5}$$

Differentiating in log space yields the local slope of the log($\mathrm{L_{H\alpha}}$) on log($\mathrm{R_{limit}}$) relation is:

$$\kappa \equiv \frac{d\log(\mathrm{L_{H\alpha}})}{d\log(R_{\rm limit})} \propto \frac{R_{\rm limit}^2}{\sigma^2} \tag{A.6}$$

Consequently, the slope κ depends on $R_{\rm limit}/\sigma$, which increases with the ratio $\Sigma_0/\Sigma_{\rm min}$. Brighter regions have larger radii before reaching the threshold. Table A.1 shows this dependence for some values of $\Sigma_0/\Sigma_{\rm min}$:

Table A.1. $R_{\rm limit}/\sigma$ and slope κ for a Gaussian profile as a function of $\Sigma_0/\Sigma_{\rm min}$.

$\Sigma_0/\Sigma_{\rm min}$	$R_{\rm limit}/\sigma$	κ	Regime
5	1.79	3.2	Very faint regions
10	2.15	4.6	Typical regions
50	2.80	7.8	Bright regions

For bright regions ($\Sigma_0/\Sigma_{\rm min} \sim 50$), $R_{\rm limit}/\sigma \approx 2,80$ and $\kappa \sim 7.8$, while for typical regions ($\Sigma_0/\Sigma_{\rm min} \sim 10$), $\mathrm{R_{limit}}/\sigma \approx 2.15$ and $\kappa \sim 4.6$. Fitting a single power law across a population spanning roughly 2 dex in $\Sigma_0/\Sigma_{\rm min}$ averages over the intrinsic curvature of the Gaussian profiles, yielding an effective slope above three, consistent with the values reported by Groves et al. (2023) and Congiu et al. (2023).

The resulting slopes for Rousseau-Nepton et al. (2018) and this work evaluated at the 5th quantile, $\kappa_{05} \sim= 0.83$ and 1.66, are consistent with a weak decrease of Σ_0 with increasing R because the method isolates the core of each region. A further consequence is that absolute-threshold methods compress the scatter perpendicular to the $L_{H\alpha}$-R relation. Regions with identical intrinsic σ but different Hα are assigned larger radii, which shifts them primarily along the correlation. This explains why the Groves et al. (2023); Congiu et al. (2023) catalogs produce a narrow distribution.

Two catalogs in our comparison achieve κ_{05} values between those of the two pure categories. In Barnes et al. (2026) the coupling between $R_{\rm cir}$ and $L_{H\alpha}$, yields $\kappa_{05} \sim 1.95$. Here, the coupling set by the absolute threshold is overcome because the flux extension is limited within each subregion of the MUSE parent mask. The resulting method is a hybrid of the two major categories of pure methods. In Zurita et al. (2026), H II regions are detected above $3\sigma_{\rm noise}$ in a background-subtracted Hα image, where the background traces the spatially variable DIG emission. This local background subtraction reduces the dynamic range of the residual image relative to the original, partially suppressing the dependence of $R_{\rm limit}$ on Σ_0 that drives κ_{05} to high values in pure absolute-threshold methods. The resulting $L_{H\alpha}$–R slope is $\kappa \sim 2.69$ for NGC 628, close to the physical expectation of $\kappa = 3$ for uniform-density regions.

Although the two methods yield intermediate κ_{05}, the mechanisms differ: in Barnes et al. (2026) suppression is topological, while in Zurita et al. (2026) it is photometric. As a result, the two approaches respond differently to DIG morphology or resolution, and neither guarantees $\kappa \approx 3$ in general.

A.2. Profile-truncated radius definition methods

In contrast, profile-truncated radii definition methods apply a relative truncation criterion that fixes $R_{\rm limit} \approx k\sigma$ independently of Σ_0, where k depends on the profile shape but not on the absolute flux level. Substituting $R_{\rm limit} = k\sigma$ into Eq. A.4:

$$L \approx 2\pi k^2\sigma^2\Sigma_0 = 2\pi R_{\rm cut}^2\Sigma_0 \tag{A.7}$$

Since $R_{\rm limit}$ is now decoupled from Σ_0, the observed slope reflects the intrinsic variation of Σ_0 with R across the population rather than the influence of the threshold:

$$\kappa = \frac{d\log(\mathrm{L_{H\alpha}})}{d\log R_{\rm limit}} = 2 + \frac{d\log\Sigma_0}{d\log R_{\rm limit}} \tag{A.8}$$

As a consequence of that, Rousseau-Nepton et al. (2018) and this work show a pronounced triangular "Christmas-tree" distribution with ~3 dex of vertical scatter at fixed R. This scatter reflects physical variations in local electron density and FF. In contrast to absolute-threshold methods, where regions with identical intrinsic σ but different Hα luminosities are assigned different radii which mainly moves the regions along the correlation rather than across it, as mentioned above.

Overall, the slope κ_{05} primarily reflects the radius-definition method when comparing different pipelines within a single galaxy. When the same method is applied across multiple galaxies, variations in κ_{05} can also reflect genuine physical differences among H II region populations.

Appendix B: Analytical derivation of the log(FF) – log(R^3) slope

The observed slope γ_R of the log(FF)-log(R) relation can be derived from the classical FF definition Eq. 7 and the $L_{H\alpha}$-R coupling effect. We confirm empirically that n_e shows no significant correlation with R across the catalogs, or it is negligible. Substituting the empirical power-law relation between $L_{H\alpha}$ and R: $\log(L_{H\alpha}) = \kappa \log(R) + \xi$ into Eq. 7 yields:

$$\log(FF) = \kappa \log(R) + \xi - 3\log(R) + \text{const} = (\kappa - 3)\log(R) + \text{const}. \tag{B.1}$$

The slope of the log(FF)-log(R) relation is therefore:

$$\gamma_R = \kappa - 3, \tag{B.2}$$

where the factor of three difference with respect to the volume arises from the change of variable $\log(R^3) = 3\log(R)$. Consequently, the slope of the log(FF)-log(R^3) relation is:

$$\gamma = \frac{\kappa - 3}{3}. \tag{B.3}$$

This result demonstrates that γ is influenced by the $L_{H\alpha}$-R slope κ, which depends on the radii definition criterion rather than on the physical properties of the ionized gas (see Sect. 6.2.1). The physical component of the observed slope is:

$$\gamma_R^{\text{phys}} = \gamma_R^{\text{obs}} - (\kappa - 3) \tag{B.4}$$

A negative γ_R^{phys} indicates that larger H II regions genuinely have lower mean rms densities, consistent with an evolutionary scenario in which more extended regions represent later stages of expansion and dispersal of the ionized gas (Sect. 7). A value of $\gamma_R^{\text{phys}} \approx 0$ indicates that the observed trend is entirely explained by the $L_{H\alpha}$-R coupling.

Appendix C: Analytical derivation of the log(FF)–log($L_{H\alpha}$) slope

The observed slope η of the log(FF)–log($L_{H\alpha}$) relation can be derived from the classical FF definition (Eq. 7) and the $L_{H\alpha}$-R coupling. Substituting the empirical power-law relation $\log(L_{H\alpha}) = \kappa \log(R) + \xi$, which gives $\log(R) = \frac{1}{\kappa}([\log(L_{H\alpha}) - \xi])$, into Eq. 7 and substituting $\log(R) = \frac{1}{\kappa}\log(L_{H\alpha}) + \text{const}$:

$$\log(FF) = \log(L_{H\alpha}) - \frac{3}{\kappa}\log(L_{H\alpha}) + \text{const} \propto \left(1 - \frac{3}{\kappa}\right)\log(L_{H\alpha}) \tag{C.1}$$

The slope of the log(FF)–log($L_{H\alpha}$) relation is therefore:

$$\eta = 1 - \frac{3}{\kappa} = \frac{\kappa - 3}{\kappa} \tag{C.2}$$

The physical component of the observed slope is:

$$\eta_R^{\text{phys}} = \eta_R^{\text{obs}} - \frac{\kappa - 3}{\kappa} \tag{C.3}$$

This result demonstrates that η is also influenced by the $L_{H\alpha}$-R slope κ, and therefore by the radii definition criterion. Three limiting cases can be considered:

Table C.1. Analytical expressions for the predicted methodological contributions to the main scaling relations. The physical component of an observed slope is obtained as $\tau^{\text{phys}} = \tau^{\text{obs}} - \tau^{\text{pred}}$.

Relation	τ^{pred}
log(FF) – log(R^3)	$\frac{\kappa - 3}{3}$
log(FF) – log(n_e)	$-2^{\dagger}$
log(FF) – log($L_{H\alpha}$)	$\frac{\kappa - 3}{\kappa}$
log(FF) – log($L_{H\alpha}/R^3$)	$1 - 2B^{\ddagger}$

Notes. † The predicted slope of the log(FF) – log(n_e) relation follows directly from the definitional identity (Eq. 7) and is independent of κ. ‡ Here, B is the slope of the empirical relation log(n_e) versus $\log(L_{H\alpha}/R^3)$ (Eq. D.6; Appendix D).

- $\kappa = 3$: $\eta = 0$. When the radii definition recovers the physical scaling $L_{H\alpha} \propto R^3$, the FF shows no methodological dependence on the luminosity-size coupling, and the full observed slope is physical.
- $\kappa < 3$: $\eta < 0$. For profile-truncated methods ($\kappa \sim 1.55$), the predicted methodological slope is $\eta \sim -0.93$. Since the observed slope is positive ($\eta^{\text{obs}} \sim +1.2$), the physical component $\eta^{\text{phys}} = \eta^{\text{obs}} - \eta^{\text{pred}} \sim +2.1$ is large and positive, reflecting the true increase of FF with luminosity in the H II region population.
- $\kappa > 3$: $\eta > 0$. For absolute-threshold methods ($\kappa \sim 4.4$-4.9), the methodological contribution is $\eta \sim +0.32$-$+0.39$, already positive, so part of the observed positive FF-$L_{H\alpha}$ correlation is a radii definition effect rather than a physical signal.

Comparing Eqs. B.2 and C.2, the three methodological slopes are related by:

$$\eta = \frac{\gamma_R}{\kappa} = \frac{\kappa - 3}{\kappa} \tag{C.4}$$

which shows that all three slopes above mentioned are influenced by the single parameter κ. The method-independent observable in slope space that absorbs this dependence is the volumetric luminosity density $L_{H\alpha}/R^3$, whose relation with FF is invariant under changes in κ, making it the method-independent observable to the induced covariance between $L_{H\alpha}$ and R, that absorbs the methodological variance introduced by different radii definitions across catalogs.

Appendix D: Analytical derivation of the log(FF)–log($L_{H\alpha}/R^3$)

Starting from the definition of the FF and the empirical log($L_{H\alpha}$)–log(R) relation with slope κ, we substitute $\log(L_{H\alpha}) = \kappa \log(R) + \xi$ into Eq. 7, obtaining:

$$\log(FF) = (\kappa - 3)\log(R) + \xi - 2\log(n_e) - 31.6 \tag{D.1}$$

We define the logarithm of the volumetric Hα luminosity density as:

$$\mathcal{E} \equiv \log(L_{H\alpha}/R^3) = \log(L_{H\alpha}) - 3\log(R) \tag{D.2}$$

which leads to:

$$\mathcal{E} = (\kappa - 3)\log(R) + \xi \tag{D.3}$$

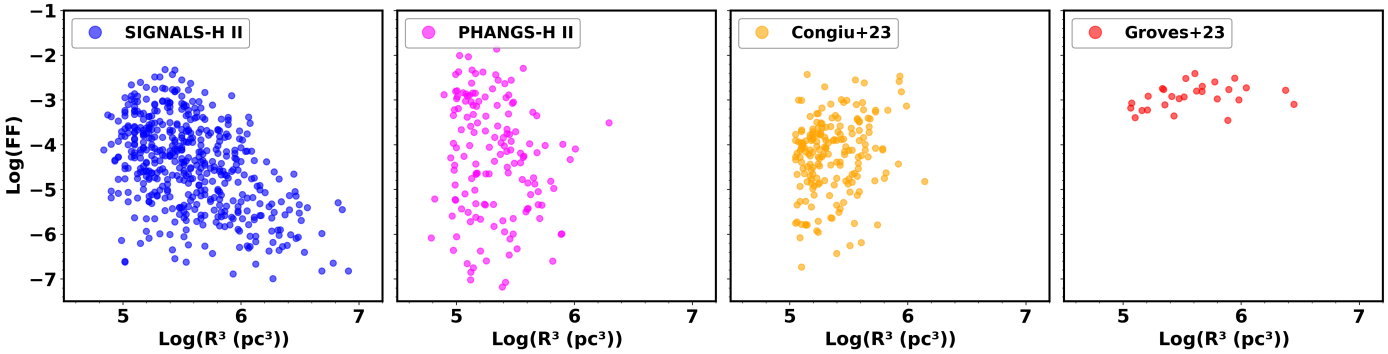


Fig. C.1. Log(FF) as a function of $\log(R^3)$ for the SIGNALS-H II (blue), PHANGS-H II (magenta), Congiu et al. (2023) (orange), and Groves et al. (2023) (red) catalogs. Catalogs adopting truncated-profile radius definitions (blue and magenta dots) show a negative trend with R, whereas absolute-threshold catalogs (orange and red dots) tend to display an elongated flatter relation, reflecting the artificial covariance between $L_{H\alpha}$ and R introduced by the threshold-based radii definition methods.

Combining Eqs. D.1 and D.3, the dependence on $(\kappa - 3)$ cancels explicitly, yielding:

$$\log(\mathrm{FF}) = \mathcal{E} - 2\log(n_e) - 31.6 = \log\left(\frac{L_{H\alpha}}{R^3}\right) - 2\log(n_e) - 31.6 \tag{D.4}$$

At fixed n_e, this implies:

$$\left.\frac{d\log(\mathrm{FF})}{d\log(L_{H\alpha}/R^3)}\right|_{n_e} = +1 \quad (\kappa \neq 3) \tag{D.5}$$

Physically, $\kappa = 3$ corresponds to the idealized case in which the mean local electron density is the same for all H II regions regardless of size, consistent with a pure Strömgren sphere model with uniform n_e across the population. In this limiting case, $\log(L_{H\alpha}/R^3)$ carries no variance across the population, and the slope of the log(FF)–$\log(L_{H\alpha}/R^3)$ relation becomes observationally indeterminate: Eq. D.5 remains valid, but it cannot be measured as a regression slope because the independent variable has zero dispersion.

When also accounting for the dependence of n_e on $L_{H\alpha}/R^3$ across the H II region population, we proceed via partial differentiation. If $\log(n_e)$ correlates with $L_{H\alpha}/R^3$ with slope B:

$$B \equiv \frac{\partial \log(n_e)}{\partial \log\left(\frac{L_{H\alpha}}{R^3}\right)} \tag{D.6}$$

The analytical expressions describing the methodological contributions to the observed slopes are summarized in Table C.1. The observed, predicted, and physical slopes recovered for each catalog are listed in Table D.1.

D.1. Slope Conditions for Method-Independent Correlations with $\log(L_{H\alpha}/R^3)$

The $L_{H\alpha}/R^3$ is a method-independent observable for characterizing the ionized gas structure across catalogs with fundamentally different radii definitions. However, this independence does not automatically extend to cross-correlations between $L_{H\alpha}/R^3$ and arbitrary photometric or physical observables. For a general observable Θ, the correlation $\log(\Theta)$-$\log(L_{H\alpha}/R^3)$ is independent of κ if and only if Θ does not depend on the region radius.

$$\left.\frac{\partial\, Log(\Theta)}{\partial\, Log(R)}\right| = 0 \tag{D.7}$$

If this condition holds, the observed correlation is purely physical and comparable across catalogs. If it does not hold, the observed slope contains a methodological contribution that scales with κ and varies between catalogs.

Observables that satisfy this condition include emission-line ratios, band ratios (R_{PAH}) and equivalent widths for example.

Observables that do not satisfy this condition include absolute luminosities in any band (L_{IR}, L_{UV}, L_{CO}), masses derived from integrated luminosities, the ionizing photons rate, the ionization parameter and the surface brightnesses for example. In the present work, the PAH-to-dust ratio R_{PAH} satisfies condition because it is a ratio of fluxes measured within apertures matched to each H II region. Its value does not depend on the H II radius, assuming that the spatial distributions of PAH emission and warm dust emission are approximately uniform within the integration aperture at the working resolution.

D.2. Comparison across catalogs

For each catalog, B is measured from a linear regression of $\log(n_e)$ on $\log(L_{H\alpha}/R^3)$. The results are summarized in the last columns of Table D.1. The values reveal that the profile-truncated catalogs (SIGNALS-H II and PHANGS-H II) yield B < 0, indicating a genuine anti-correlation between local electron density and volumetric emissivity. Thus H II regions with higher $L_{H\alpha}/R^3$ tend to have slightly lower n_e, consistent with the luminous regions hosting large ionized volumes at moderate density rather than compact high-density gas. In contrast, Congiu et al. (2023); Groves et al. (2023) and Barnes et al. (2026) yield positive and B ~0. As noted in Subsubsect. 5.4.1, Barnes et al. (2026) derive electron densities from the Groves et al. (2023) catalog, so the two catalogs share the same n_e distribution. The compressed dynamic range of n_e in these samples (Fig. 4) suppresses the statistical detectability of the physical anti-correlation, driving B ~0 regardless of whether the physical relation exists. Congiu et al. (2023), which also uses an absolute-threshold segmentation method but a wider range of electron densities, yields B much lower than zero because B depends on the dynamic range of n_e in each catalog.

The physical component of the observed slope from the log(FF)–$\log(L_{H\alpha}/R^3)$ relation is:

$$\eta^{\mathrm{phys}}_{L/R^3} = \eta^{\mathrm{obs}}_{L/R^3} - 1 \tag{D.8}$$

Across all three catalogs with large dynamic range of n_e (SIGNALS-H II, PHANGS-H II and Congiu et al. (2023)), the

Table D.1. Observed (τ^{obs}), predicted (τ^{pred}), and physical ($\tau^{phys} = \tau^{obs} - \tau^{pred}$) slopes for the main scaling relations in the five H II region catalogs.

Catalog	κ_{05}	log(FF) − log(R^3)			log(FF) − log($L_{H\alpha}$)			log(FF) − log($L_{H\alpha}/R^3$)			
		obs	pred	phys	obs	pred	phys	obs	$B^{\dagger}$	pred††	phys
SIGNALS-MUSE	0.83	−1.13	−0.72	−0.41	1.60	−2.61	4.21	1.294	−0.138	1	0.294
PHANGS-H II	1.66	−1.18	−0.45	−0.73	1.22	−0.81	2.03	1.303	−0.160	1	0.303
Barnes et al. (2026)	1.95	−0.32	−0.35	+0.03	−0.006	−0.54	+0.53	0.978	0.217	1	−0.022
Congiu et al. (2023)	4.53	0.77	+0.51	+0.26	1.10	+0.34	+0.76	1.296	−0.148	1	0.296
Groves et al. (2023)	3.95	0.23	+0.32	−0.09	0.31	+0.24	+0.07	0.931	0.035	1	−0.069

Notes. Observed slopes of the log(FF) − log(R^3) and log(FF) − log($L_{H\alpha}$) relations are described in Appendices B and C, respectively. † B is the slope of the empirical relation log(n_e) versus $L_{H\alpha}/R^3$ measured independently for each catalog (Appendix D), and is listed for reference only. †† For the relation log(FF)-$L_{H\alpha}/R^3$, the predicted slope is 1 without accounting for the dependence of n_e on $L_{H\alpha}/R^3$ (Appendix D). Predicted slopes are computed from the analytical expressions summarized in Table C.1.

physical slopes in the last "phys"' column of Table D.1, once the predicted slope of +1 is accounted for, reflect the true physical signal whereby more luminous H II regions are associated with higher values of log(FF). This confirms that log($L_{H\alpha}/R^3$) provides a complete and quantitative account of the slope variations across catalogs with fundamentally different radius definition methods.

Note that the observational slopes derived here have been obtained from a univariate analysis. To account for the nonzero values of B, a multivariate regression of log(FF) on log($L_{H\alpha}/R^3$) and log(n_e) simultaneously would be required.

Appendix E: Effect of the galactocentric luminosity correction

The radial luminosity correction was calculated but only appears as significant for the SIGNALS catalog which removes the decrease of $L_{H\alpha}$ with galactocentric radius, characterized by a gradient of $\lambda = -0.56 \pm 0.03$, supported by the regression analysis and Spearman's rank correlation ($\rho = -0.38$, $p < 0.001$).

To assess the impact of this procedure, we repeated the analysis using the corrected luminosities (Table E.1). The correction has a negligible effect on the log(FF)–log(R^3) and log(FF)–log(n_e) relations in SIGNALS-H II, with median differences of only 0.05 and 0.01 dex, respectively, well within the corresponding 1σ uncertainties. A somewhat larger difference of 0.12 dex is found for the log(FF)–log($L_{H\alpha}$) relation, which is expected because $L_{H\alpha}$ enters explicitly as the independent variable.

Table E.1. Effect of the galactocentric luminosity correction on the slopes of the FF scaling relations for the SIGNALS-H II sample.

Relation	With correction	Without correction	Δ
log(FF)–log(R^3)	−1.18 ± 0.10	−1.13 ± 0.10	0.05
log(FF)–log(n_e)	−2.64 ± 0.10	−2.65 ± 0.10	0.01
log(FF)–log($L_{H\alpha}$)	1.72 ± 0.06	1.60 ± 0.06	0.12

Notes. Slopes and 1σ uncertainties from ordinary least-squares regression.

Importantly, the galactocentric correction does not modify the inferred $L_{H\alpha}$–R scaling exponents in SIGNALS, including the lower-envelope estimator κ_{05} used throughout this work.

Appendix F: Selection criteria for H II regions

In Table F.1, we summarize the selection criteria applied to identify the final H II regions in the SIGNALS-H II and PHANGS-H II catalogs. These criteria include constraints on emission line fluxes, signal-to-noise ratios, pseudo-Voigt profile parameters, and classical BPT diagnostics.

Appendix G: Additional figures

Table F.1. Criteria for H II region selection

Column	Unit	Description	Criteria
		SIGNALS survey	
R_g	kpc	Galactocentric radius	deprojected with i of Lang et al. (2020)
$L_{H\alpha}$	10^{-7} erg s^{-1}	Hα total luminosity corrected for extinction	$\neq 0$
I0p	mW/m^2	pseudo-Voigt profile parameter I0	$\neq 0$
sigp	pc	pseudo-Voigt profile parameter σ	$\neq 0$
R2p		pseudo-Voigt profile correlation coefficient R^2	0.65
log(NII/Hα)	Log	Log line ratio [NII]6583/Hα	BPT diagnosis criteria
SNR1		line ratio [NII]6583/Hα best S/N	>4
log([OIII]/Hβ)	Log	Log line ratio [OIII]5007/Hβ	BPT diagnosis criteria
SNR4		line ratio [OIII]5007/Hβ best S/N	>4
log([SII]16/[SII]31)	Log	Log line ratio [SII]6716/[SII]6731	Log([S II] 6716/6731) $\in$ (-0.36, 0.137)
SNR10		line ratio [SII]6716/[SII]6731 best S/N	> 4
log([SII]6716 + 6731/Hα)		Log line ratio [SII]6716 + 6731/Hα	$\geq$ 0.4
SNR2		line ratio [SII]6716 + 6731/Hα best S/N	> 4
		PHANGS-MUSE Ionized Regions Catalog	
pseudo-Voigt R2		pseudo-Voigt profile correlation coefficient R^2	0.65
Emission line fluxes			>0
EW H1 6563A		Hα equivalent width	< 0
S/N H1 6563A		line ratio Hα best S/N	>4
S/N H1 4861A		line ratio Hβ best S/N	>4
S/N N2 6583A		line ratio [NII]6583/Hα best S/N	>4
S/N O3 5007A		line ratio [OIII]5007/Hβ best S/N	>4
S/N S2 6716A		line ratio [SII]6716/ best S/N	>4
S/N S2 6731A		line ratio [SII]6731/ best S/N	>4
log([SII]16/[SII]31)		Log line ratio [SII]6716/[SII]6731	Log([S II] 6716/6731) $\in$ (-0.36, 0.14)

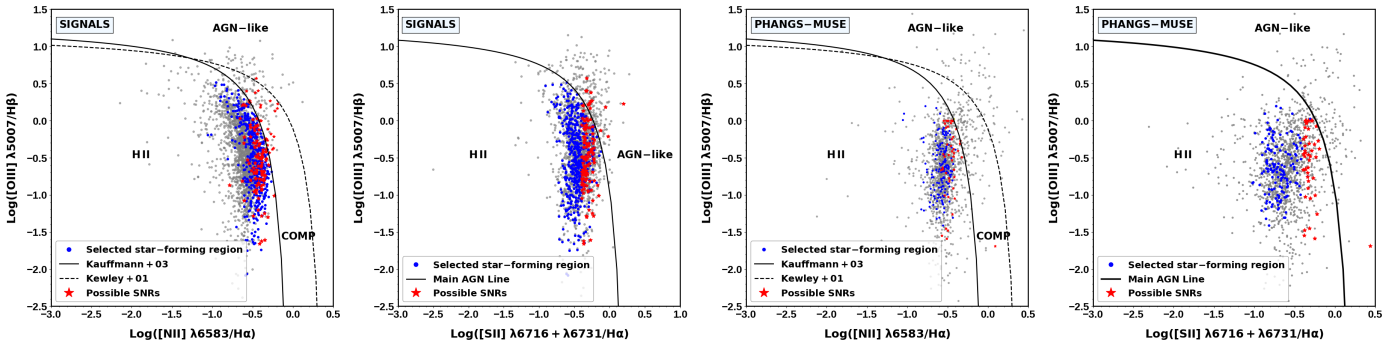


Fig. G.1. BPT-NII and BPT-SII diagrams of the SIGNALS and PHANGS sample in NGC 628. In both panels, the solid and dashed black lines indicate the theoretical boundaries separating the domain of the star-forming, composite and AGN-like regions as prescribed by Kewley et al. (2001) and Kauffmann et al. (2003). The upper and lower panels show the SIGNAL-H II and PHANGS-H II regions, respectively, represented as blue points. Regions identified as possible supernova remnants (SNR) are included as red stars. Gray points indicate regions with $R^2_{fit} > 0.65$.

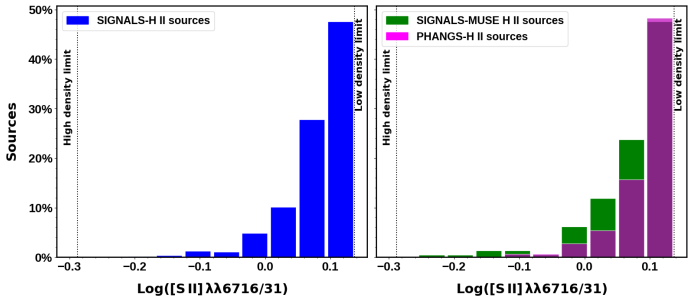


Fig. G.2. In situ electron density, n_e, against the ratio [S II]6716/[S II]6731 in NGC 628 for the SIGNALS-H II subsample (left panel). In the right panel, we compared PHANGS-H II regions with SIGNALS-MUSE, a subsample from SIGNALS-H II within the MUSE mask FoV. Vertical dashed lines indicate, from left to right, the high and low electron density limits based on our criteria outlined in Subsubsect. 5.4.1.

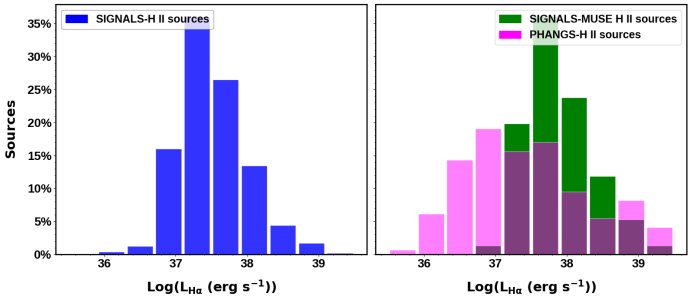


Fig. G.3. Hα luminosity distribution of the H II regions selected in our final sample. The left panel focuses only on the SIGNALS-H II subsample. In the right panel, we compared the PHANGS-H II subsample (magenta) with the restricted SIGNALS-MUSE sample (cyan). The vertical axis represents the total number of regions in percentage.

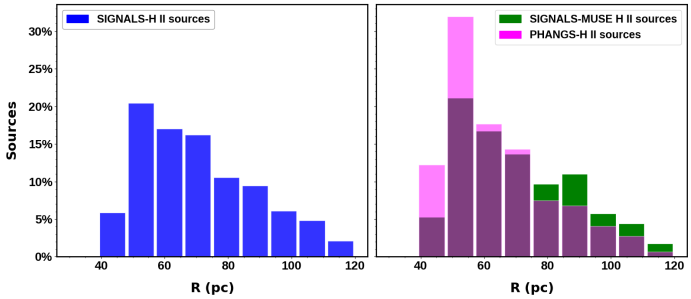


Fig. G.4. Size distribution of the radii of the H II regions in our final sample considering only the SIGNALS-H II subsample (left panel). In the right panel, we compared the PHANGS-H II subsample (magenta) with the restricted SIGNALS-MUSE sample (cyan). The equivalent radii of the PHANGS-H II subsample are the radii obtained by circularizing their ellipsoidal volume. The vertical axis represents the total number of regions in percentage.

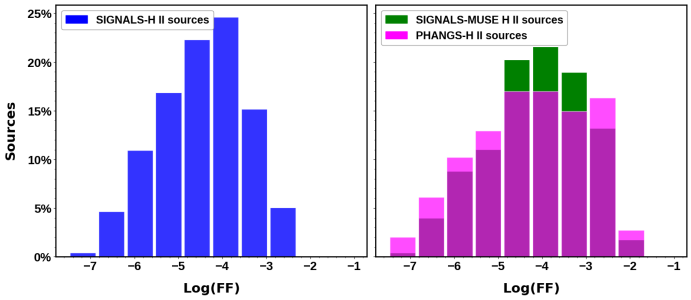


Fig. G.5. FF distribution in our final sample considering the SIGNALS-H II subsample (left panel), and compared the PHANGS-H II subsample (magenta) with the restricted SIGNALS-MUSE sample (green) in the right panel. The vertical axis represents the total number of regions in percentage.

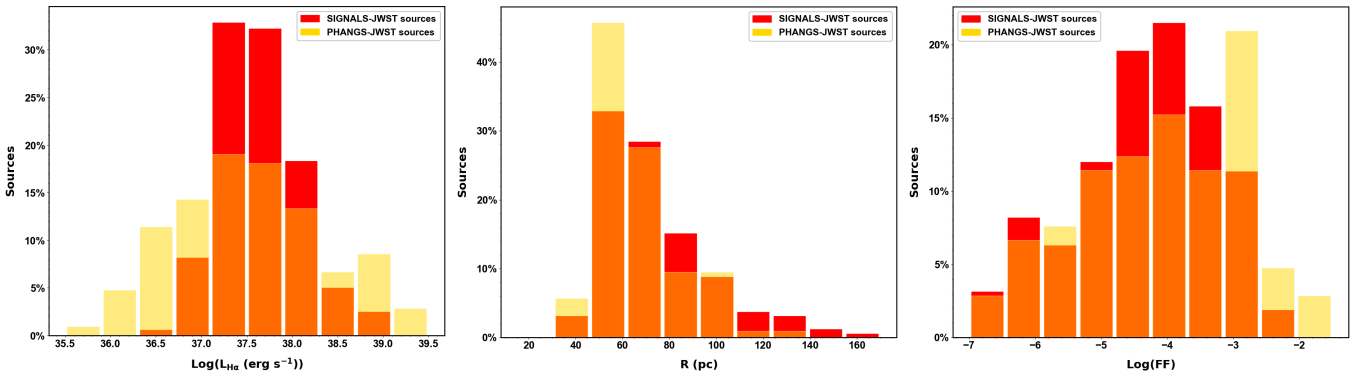


Fig. G.6. Histograms showing the Hα luminosity distribution (left panel), the size distribution of the circularized radii (central panel), and the number of regions as a function of the FF (right panel) for SIGNALS-JWST sample (yellow) and PHANGS-JWST sample (red) (Sect. 6.4). The vertical axis represents the total number of regions in percentage.